Strategy and Coordination in Risky Household Decisions

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Abstract

Do spouses successfully coordinate risk-taking decisions within the household? This paper uses a lab-in-the-field experiment with married couples in Bangladesh to show that most people hold mistaken beliefs about their spouses' risk-taking behavior. In a sequential lottery-choice game, husbands and wives each selected a lottery while sharing the combined winnings from both. Under imperfect information about their spouse's choice, only one in four subjects successfully coordinates their choice with their spouse's so as to achieve their intended household risk-exposure level. Biased beliefs about the spouse's choice led 23% of subjects to accept excessively risky lotteries and 24% to sacrifice profitable opportunities. Coordination failures are particularly pronounced when one spouse actively attempts to counter or ignore the other's choice, a behavior typically shown by men. Measures of women's bargaining power and non-cognitive skills help explain these gender differences. Coordination errors in the experiment predict biased beliefs about a spouse's savings behavior for different purposes, such as investments in children and emergency needs. Crucially, the size of the coordination error is positively associated with the likelihood that a household was unable to cope with adverse shocks in the past twelve months using available resources, forcing households to sell productive assets or resort to migration. *Keywords:* Risk preferences, intrahousehold decision-making, gender, spousal cooperation, lab-in-field experiment, rural households, Bangladesh

JEL Codes: C93, D13, D81, J16, O12

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1 Introduction

A household's risk exposure is the result of decisions made by each of its members. Most standard models of household decision-making assume a unitary decision-maker following Becker's framework (Becker, 1981), or assume Pareto-efficient outcomes will arise from intrahousehold bargaining (Chiappori, 1992, 1997). In practice, divergent priorities and imperfect information about one another's decisions can lead to coordination failures between partners. Poor coordination can expose a household to excessive risk if a household member makes risky decisions under the false impression that their spouse has sufficient assets to cover them in case of a crisis. Or, it can cause households to sacrifice legitimate investment opportunities because both spouses are overly conservative.

This paper applies an experimental lens to intrahousehold risk-taking decisions. Precisely, it examines whether spouses coordinate risk-taking even when one partner's actions are not observable. I test possible reasons for coordination failure, namely, divergence in spousal risk preferences and willingness to accommodate spouses' preferences. To this end, I use a lab-in-the-field design that artificially creates asymmetric information between couples to mimic real-life situations where one spouse is not fully informed about the other's decisions, even though both derive utility from the joint outcome. Such dynamics arise from delegation in decision-making or, in patriarchal settings, gender-based task specialization.

The experiment takes place in Bogura, a mostly agricultural district in Northern Bangladesh. Married couples from fruit and vegetable producing households played a sequential lottery choice game[1]. Couples in my sample segregate decision-making: women focus on children's education, healthcare, and animal rearing, while men handle all farming decisions. In the experiment, each spouse chose a lottery from a common set of lotteries. The first-mover made their choice without knowing how their spouse, the second-mover, would respond. The second-mover was allowed to state their choice conditional on the first-mover's choice. To minimize the chance of conflict, the first-mover's choice was not divulged to the second-mover; rather, the second-mover stated their preferred choice for every possible first-mover action. This setup also lets me observe how individuals respond to changes in their spouse's choices. First-movers predicted their spouse's conditional choices, allowing measurement of belief biases and deviations between expected and actual household portfolios.

I reject the hypothesis that first-movers can achieve their intended household risk-exposure level. Only 27% of men and 22% of women second-movers choose the lottery their spouse expected. Comparing achieved household exposure with first movers' preferred risk levels reveals similarly large discrepancies with the preferred risk exposure level. Both spousal risk preference disparity and the second-mover's willingness to accommodate their spouse's choice affect the divergence of household risk exposure from the intended outcome. Gendered differences arise in how second-movers react to the first-mover's lottery choice. Women were far more accommodating of the first-mover's lottery choice than men, which is reflected in a greater deviation between household risk exposure and women's expectations than men's.

[1]The experiment was pre-registered in the AEA RCT Registry (AEARCTR-0014589).

More risk averse women and women who work for a living show a greater tendency to counter the first-mover's lottery choice. Whereas men whose wives have access to their independent earnings show a greater tendency to counter their wives' choices. While an independent income source by improving outside options could increase bargaining powers for women (Chiappori, 1992, 1997), the association between men's strategy and women's employment is indicative of non-cooperative models of bargaining, like the separate spheres model (Lundberg and Pollak, 1993), if as a result of low cooperation between spouses, each retreats to their own domains in the household. Non-cognitive skills or psychological personality traits also explain how men and women respond to the first-mover. Women with higher *extraversion* and lower *conscientiousness* are more deliberate in their choice of lottery and more likely to counter the spouse. Among men, those who are more *open to new experiences* are better at accommodating their wives' preferences.

I use household data on partners' assets, liabilities, and behavior under extraneous stress to validate the experimental findings. I find evidence of considerable informational asymmetry between spouses. Nearly 40% of the women and 35% of the men had incorrect beliefs about whether their spouse had made any savings in the past year. The mistakes also extend to physical assets and liabilities. The size of the first-mover's prediction error in the game is positively associated with the likelihood of mistaken beliefs about their partner's pecuniary and physical assets. Misunderstandings about a partner's risk exposure can increase the vulnerability of both partners' ventures during crises. The size of the coordination error in the game predicts the likelihood of unplanned sales of productive assets and migration to cope with an adverse shock, such as a drought, in the past year. These findings should be interpreted as suggestive, as data limitations preclude the determination of causality.

This study bridges two strands of economic literature: the literature on strategic interactions in household decision-making and the literature on portfolio choice and risk-sharing. The earliest literature on risk-sharing and portfolio choice in households is attributed to Mazzocco (2004), Mazzocco and Saini (2012) and Browning and Chiappori (1998). Risk-sharing between partners is achieved through transfers between partners that insulate the expenditures of a partner affected by a shock. Most studies use observational data to validate the risk-sharing hypothesis. This experiment examines portfolio choices between partners under conditions in which household specialization creates the potential for hidden actions. The results, which reveal widespread deviation from one partner's intended household portfolio, should be seen as a precursor to decisions about resource sharing between partners in distressed households. The lack of coordination in these decisions is reflected in households affected by these decisions being more likely to have experienced the liquidation of productive assets during recent periods of distress.

One of the few experimental worksin this area is Schaner (2015), whose field experiment among Kenyan couples, shows that disparities in time preferences lead to inefficient saving decisions that sacrifice higher returns for greater individual control over savings. The present study confirms this finding for divergence in risk preferences and is generalizable to all household decisions involving uncertain outcomes. The experiment deliberately eschews contextualizing the decision as a financial savings or investment decision, so as to make the findings more generalizable to any household decision that involves risk, such as farm input decisions, children's education, or the purchase of physical assets. In South Asian societies, financial decisions are largely controlled by men, but even in patriarchal societies, women make essential decisions about aspects like investments into children or, in the context of Bangladesh, animal rearing.

Recent literature on household risk behavior also account for behavioral elements such as social norms in portfolio decisions (Guha Thakurta and Wesselbaum, 2026a). In contrast to Guha Thakurta and Wesselbaum (2026a), this paper's findings suggest that women's bargaining power influences risk-taking decisions, as household portfolios reflect women's choices more, though at the cost of lower coordination between partners. In this respect, this study relates to works of Conlon et al. (2021) and Abbink et al. (2020), which reveal gender differences in how partners value each other's knowledge and accommodate each other's agency in decision-making in South Asian societies. By allowing for imperfect information about a spouse's actions, I observe how gender differences in how people react to signals from their spouse lead to household outcomes that more closely reflect men's preferences. Going further, I show that traditional determinants of bargaining power and non-cognitive skills can explain these gender differences.

In addition to contributing to our understanding of household decision-making, these findings offer potential insights for policy on female agency in the household. Beyond what is accepted regarding the importance of female employment for more equitable household outcomes, this study demonstrates the impact of non-cognitive skills on outcomes. The limitations of a cross-sectional design prevent me from separating the effect of education or upbringing from that of non-cognitive skills. However, it is important that personality traits, such as extraversion among women and openness to new experiences among men, have an independent association with greater representation of independent choices by women, even after controlling for education. This is an avenue policymakers should consider more carefully in designing development strategies and measures to increase women's empowerment.

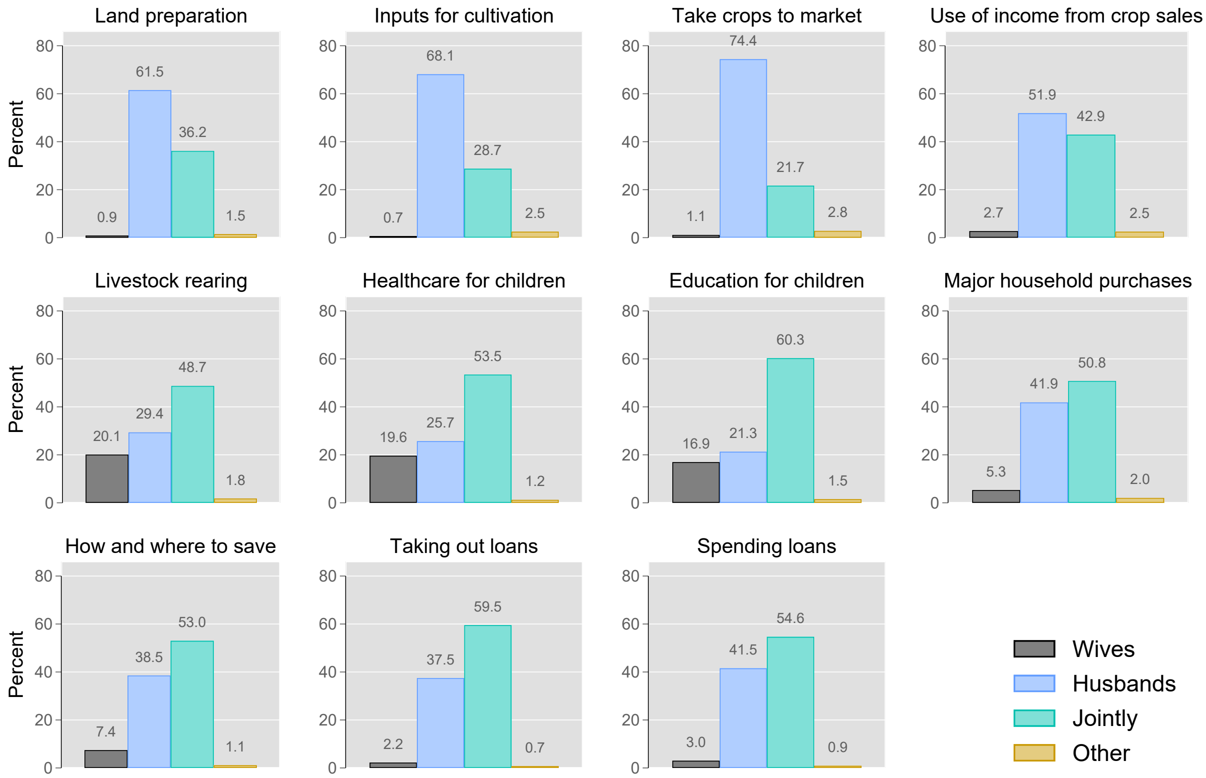
2 Setting and Sample Characteristics

The experiment was conducted using married couples recruited from forty-six villages in the Bogura district of Bangladesh. These households were recruited as part of a larger study examining the demand for cold storage facilities among fruit and vegetable growers in Bangladesh. The experiment was conducted by enumerators at the participants' homes during the baseline survey for the larger study. The survey was conducted in November and December of 2024. 1059 households from 46 villages in two upazillas (Shibgonj and Kahalu) of Bogura district were surveyed as part of the baseline, while the experiment was conducted for 1774 individuals from 887 of the households. The experiment was not conducted if either respondent refused to participate or an eligible household member was not present when the enumerators visited. Among the households where the experiment was conducted, data on women's household savings were missing for two. I exclude these two households from the final sample, which is composed of 1770 individuals or 885 couples[2]. The non-response rates are well within the bounds expected during power calculations. Power calculations were based on figures obtained from earlier related studies[3].

[2]Non-participant men had a higher proportion of individuals with less than primary education, which could indicate more traditional-minded individuals compared to participants (Appendix Table B.1).

[3]I relied on Chowdhury et al. (2022) for sample characteristics such as standard deviations in the outcome and probable effect size. The design allows detection of a 0.2 standard deviation size effect with at least 0.8 power.

Figure 1: Distribution of decision-making powers in sample households as reported by women.



Bangladesh offers an interesting backdrop for testing cooperation between spouses in household decisions. Most unions are long-lasting since divorce is not very common [4]. Thus, unions may persist despite growing dissonance between partners. Within the household, tasks are assigned to husband and wife based on existing gender norms (Choudhury, 2019). The patriarchal nature of Bangladeshi society means that most decision-making powers are vested in the husband. Since the resulting resource distribution may be very unequal, women have an incentive to engage in non-cooperative behavior (Anderson et al., 2018). The exclusion of women from important spheres of household decision-making also increases the likelihood of asymmetric information between spouses. There are also several spheres of household decision-making where women play a significant role. In our sample, women were most empowered to make decisions regarding the education and health care of children, as well as the rearing of livestock. Figure 1 shows how decision-making powers are divided in households between men and women. While most men have sole decision-making powers in spheres of household activity which take place outside the household (farming decisions like cultivation decisions and when to take crops to the market), most women have sole or joint decision-making powers in the spheres of household activity which take place inside it (decisions about children's health care, education and purchases for the household). Importantly

[4]The divorce rate in Bangladesh was 1.1% in 2023 as per figures released by the Bangladesh Bureau of Statistics (The Business Standard, 2024).

in decisions regarding household savings and credit, most women reported having either sole or joint say.

Within Bangladesh, the choice of Bogura district was motivated by the needs of the larger study assessing the need for cold storage for fruits and vegetables. Bogura is one of the major vegetable-producing regions in Bangladesh (Mou et al., 2019). It has a largely rural population with a literacy rate slightly below the national average (Bangladesh Bureau of Statistics, 2024). Our sample is representative of agricultural communities in Bangladesh and similar to samples used in earlier studies on related topics (Abbink et al., 2020; Chowdhury et al., 2022).

Table 1: Demographic characteristics

	Households		Men		Women	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
	(1)		(2)		(3)	
Muslim (%)	97.30	(16.25)				
Dependency ratio[a]	0.25	(0.19)				
Number of assets:[b]						
Mobile phone	2.42	(1.10)				
Television	0.78	(0.55)				
Cow/Buffalo	1.22	(1.53)				
Poultry	15.63	(14.50)				
Lives in household:						
Child under 5 years (%)	37.51	(48.44)				
Husband's mother (%)	20.00	(40.02)				
Husband's father (%)	11.98	(32.49)				
Wife's mother (%)	2.82	(16.58)				
Wife's father (%)	0.45	(6.71)				
Age (years)			42.69	(10.93)	35.39	(10.33)
Household head (%)			95.48	(20.79)	0.00	(0.00)
Spouse of household head (%)			0.00	(0.00)	95.48	(20.79)
Education:						
Less than primary (%)			15.03	(35.76)	19.32	(39.50)
Primary (%)			32.54	(46.88)	28.25	(45.05)
Secondary (%)			34.12	(47.44)	41.47	(49.30)
Higher-secondary or more (%)			18.30	(38.69)	10.96	(31.26)
Main occupation:						
Ag. self-employed (%)			78.53	(41.08)	4.07	(19.76)
Non-ag. self-employed (%)			11.19	(31.54)	1.13	(10.58)
Ag. wage labor (%)			1.58	(12.48)	0.68	(8.21)
Non-ag. wage labor (%)			6.78	(25.15)	1.36	(11.57)
Livestock rearing (%)			0.00	(0.00)	7.68	(26.65)
At home (%)			0.00	(0.00)	84.29	(36.41)
Other (%)			1.92	(13.73)	0.79	(8.86)
Observations	885		885		885	

Notes: Standard deviation in parentheses for mean outcomes.

[a] Dependency ratio was calculated as the ratio between the number of household members not of working age (below 15 and above 65 years) and the number of household members of working age (15 to 65 years).

[b] These numbers are as reported by the adult male member. There were minor differences between the numbers reported by men and women.

Table 1 presents summary statistics of some household and individual demographic characteristics. Our sample was primarily composed of low-income Muslim households. Around 40% of the households had a young child. The average man was around 40 years of age and the average women 35. In nearly all of the cases the adult male in the couples recruited headed their own household. Due to patrilocal marital arrangements, it is more common for the husband's parents to be co-living them than the wife's parents. 85% of men and 81% of women had completed at least primary education. Most men worked on own their farms (78%) while most women were homemakers (84%). The institution of *purdah* which is still practiced in Bangladesh enforces women's exclusion from public spaces restricting their choice of occupation. Still, the second-most common primary occupation for women was livestock rearing (8%).

3 The Experiment

3.1 Design

Wives and husbands were interviewed in separate locations of their household by interviewers of the same gender, to minimize interactions during gameplay. Players were given the game after answering some questions about socioeconomic conditions[5]. Enumerators used tablet computers loaded with survey instruments developed using the *SurveyCTO* platform. Each husband-wife pair participated in multiple games. I will concentrate on the findings from two of these games in this paper. Since players were only paid for their performance in one randomly selected game, a player's performance in one game should not affect their decisions in another game [6].

After a player and their spouse were led into their respective room, enumerators explained the rules of the game. Each spouse was presented with the list of the lotteries in Table 2. Each lottery had two equally likely outcomes, and variation in perceived riskiness was created by varying the size of the potential outcomes. The lottery number (LN) in column (1) indexes the lotteries from least risky to most risky based on how risk-averse a person would have to be to choose the lottery from the available six options[7]. Players were informed that they and their spouse would each take turns choosing a lottery from the list and that the combined winnings from the two lotteries would be equally divided between them. This meant that each player's winnings and their welfare were dependent on both their own and their spouse's choice. The lottery choices were made sequentially:

1. First, the player in Room A (in Figure 2), the *first-mover*, chose a lottery without any information about their spouse's choice.
2. Then the player in Room B, the *second-mover*, made their choice *conditional* on the first-mover's choice.

Table 2: Set of alternative lotteries presented to players, with corresponding risk levels.

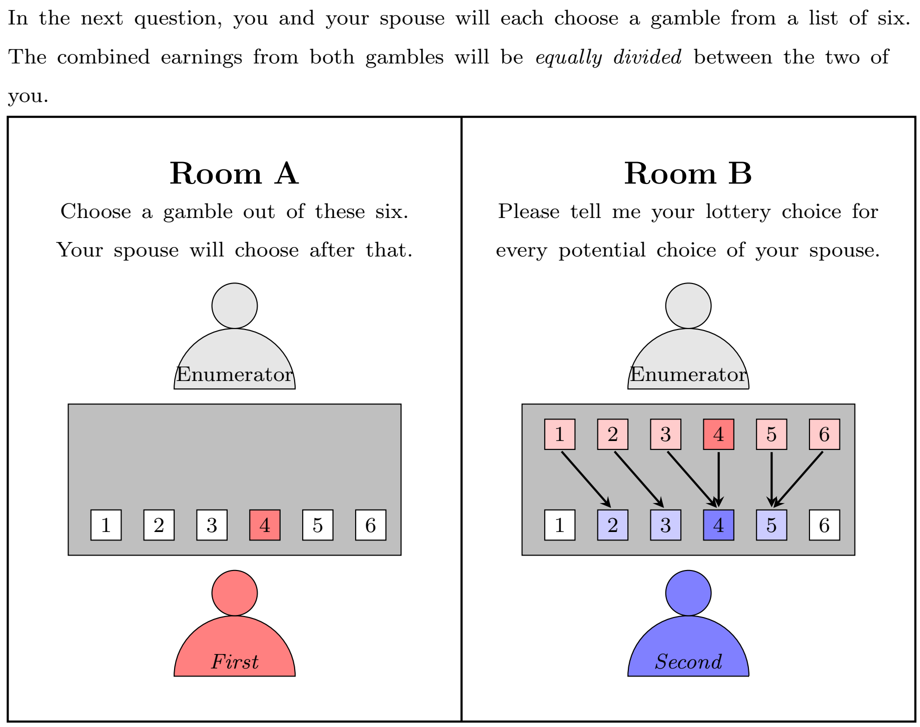
Lottery Number (1)	Payoffs (BDT) (2)	Gamma (γ) Range (3)	Gamma (γ) Midpoint (4)	Risk Aversion (5)
1	125 and 125	$[3.03, +\infty)$	3.03	Extremely averse
2	190 and 100	$[1.99, 3.03)$	2.51	Very averse
3	240 and 90	$[0.60, 1.99)$	1.30	Moderately averse
4	350 and 45	$[0.20, 0.60)$	0.40	Slightly averse
5	400 and 15	$[0.12, 0.20)$	0.16	Risk-neutral
6	450 and 0	$(-\infty, 0.12)$	0.12	Risk-seeking

[5] Men were additionally answered some questions about fruits and vegetables production and storage methods in the household.

[6] Appendix A details the designs of each game or treatment.

[7] This is done assuming a decision-maker with a utility function that satisfies Constant Relative Risk Aversion (CRRA).

Figure 2: Example of sequential lottery choices.



To minimize the threat of conflict between couples post-game play, the *second-mover*'s choice was elicited using the strategy method. The *first-mover*'s actual choice was kept secret from their spouse; *second-movers* were instead asked to dictate their lottery choice for all possible choices the *first-mover* might make. Having players think over all the possible choices of their spouse also helped ensure that individuals were deliberate in their decisions (Charness et al., 2012). The *second-mover*'s strategy, so elicited, is used to characterize their responsiveness to the spouse's choice. To identify outcome deviation from each player's expected or intended risk exposure for the household, *First-movers* were incentivized to predict their spouse's strategy as *second-mover* [8].

Each couple participated in two rounds of this game, once with the wife playing as the *first-mover* and the husband as the *second-mover*, and again with their roles switched. At the end of all the games, the enumerator revealed only the outcomes of the lotteries selected by the player, not those of their spouse. Each household member was handed their share from the winnings without revealing which round they were being paid for.

3.2 Prediction errors

The first-mover's prediction about the second-mover's lottery choice is used to measure how far the second-mover's choice deviates from the first-mover's expectation. *Error* is defined

[8] First-movers were paid an additional 150 Taka if they could correctly predict their spouse's strategy, 100 Taka if they got only one lottery choice wrong or 50 Taka if they got two lottery choices wrong. This additional amount is paid out only in the instance that that specific game is selected for payment.

as the difference between the first-mover's prediction and the second-mover's actual choice. Lottery choices are indexed using the lottery number (LN).

$$Error = E_1(LN_2) - LN_2 \tag{1}$$

where LN_2 is the second-mover's chosen lottery number and $E_1(LN_2)$ is the first-mover's prediction about the second-mover's choice. Since both the second-mover's entire strategy and the first-mover's prediction of that strategy were elicited, it is possible to measure prediction deviation over the strategy. However, unless otherwise stated, *Error* is meant to denote the deviation for the actual choice of the second-mover.

Error is zero only if the first-mover correctly predicts the second-mover's choice, it is negative if the first-mover overestimates how safe the second-mover's choice is, and positive if the first-mover underestimates how safe it is. It take integer values in the interval $[-5, 5]$. *Abs. error* is constructed as the absolute value of *Error* and it takes integer values in the interval $[0, 5]$. Non-zero values of *Abs. error* indicates that the second mover's choice deviates from the first-mover's expectation.

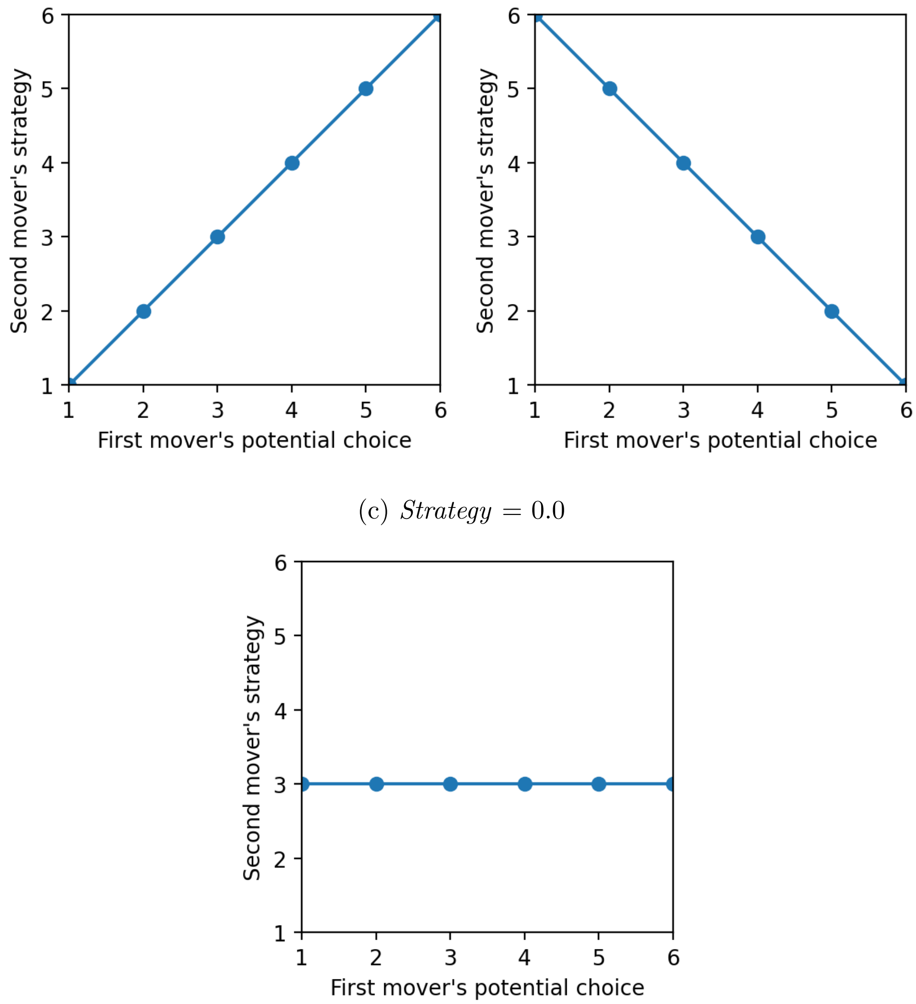
3.3 Strategy of second-mover

A second-mover states their strategy in the *Joint* game as the lottery that they would play if the first-mover plays lottery 1 or if they play lottery 2, and so on. Using the lottery number (LN) to index the second-mover's lottery choices in the vertical axis and the first-mover's potential choice in the horizontal axis, sub-figures 3a, 3b, and 3c illustrate three examples of strategies actually followed by second-movers. These represent three extremes. Sub-figure 3a represents 163 second-movers who decided that they would play the exact same lottery that their spouse chose as the first-mover. Sub-figure 3b represents the 6 subjects whose strategy was to play the most risky lottery if the first-mover played the safest, to play the second-most risky if the first-mover played the second safest, and so on. Sub-figure 3c represents the 4 subjects who chose to play lottery 3, no matter what the first-mover played. The *Strategy* variable is calculated as the correlation coefficient between the second-mover's choice of lottery (vertical axis) and the first-mover's potential choice. Hence, the value of the *Strategy* variable for panel (a) is 1.0, for panel (b) is -1.0, and for panel (c) is 0.0. Most second-movers *Strategy* lie somewhere between these values.

Figure 3: Examples of strategies followed by second-movers.

(a) *Strategy* = 1.0 (b) *Strategy* = - 1.0

(c) *Strategy* = 0.0



Notes: The lottery number (LN) indexes the second-mover's lottery choices in the vertical axis and the first-mover's potential choice in the horizontal axis.

3.4 Individual risk preferences

Individual risk preferences were elicited using the Binswanger (1981) method. This was done before the games described above. Each subject was asked to choose one lottery from the lottery list in Table 2. Based on their choice, the lottery number (LN) characterized their risk preference. To make the incentive structure similar to that in the game, players' winnings were shared equally with their spouses. The Gamma (γ) values in column 4 of the table represent the risk-aversion parameter, which satisfies the player's observed choice under constant relative risk aversion. Unless otherwise indicated, individual risk preferences are expressed in terms of the lottery number (LN) and preference disparity between spouses as the difference between the player's lottery number and their spouse's lottery number.

Appendix A provides more details on the calculation.

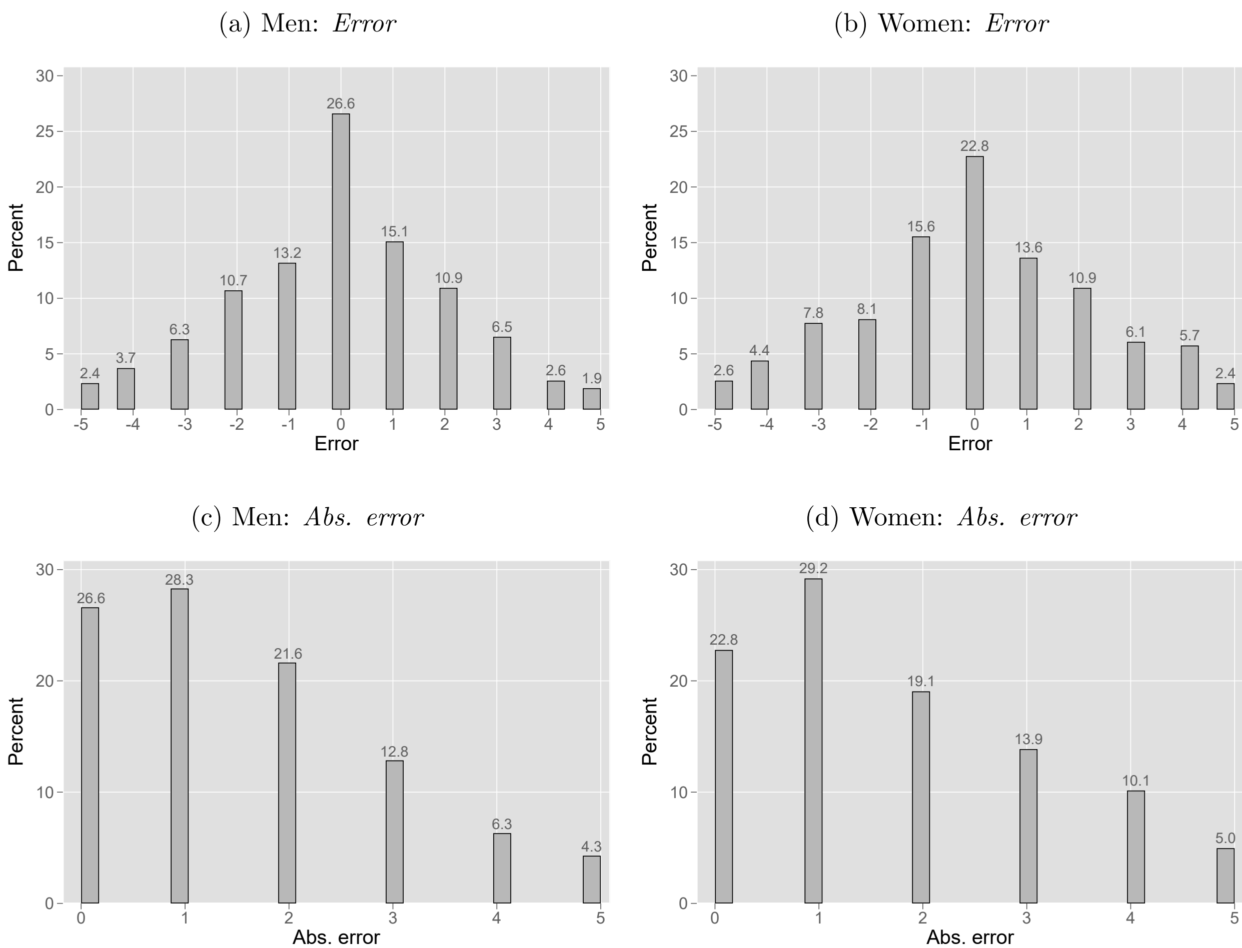
3.5 Non-cognitive skills

Non-cognitive skills or personality traits were measured using a 15-item Big Five questionnaire derived from Chowdhury et al. (2022). The five-factor module (Big Five) is widely used in economic research and is the standard method in most longitudinal datasets (Vella, 2024). The Big Five taxonomy proposes five dimensions of personality, namely, *openness to experience* (ability to be creative, curious, intellectually engaged, honest/humble, and inquisitive), *conscientiousness* (self-discipline, punctuality, and organized and general competence), *extraversion* (how talkative, friendly, energetic, and outgoing the person is), *agreeableness* (the tendency to be kind, charitable, warm, and generous), and *neuroticism* (fear, worry, paranoia, and stress) (Vella, 2024).

4 Results and Discussion

Spouses do not achieve their intended risk exposure level for the household, as only a quarter of second-movers actually choose the lottery number expected by the first-mover. 27% of women second-movers choose the lottery number expected by their husbands, and 23% of men second-movers choose the lottery number expected by their wives[9]. Figure 4 presents the distribution of the (a) *Error* for men first-movers, (b) *Error* for women first-movers, (c) *Abs. error* for men first-movers, and (d) *Abs. error* for women first-movers.

Figure 4: Distributions of the *Error* and *Abs. error* variables for men and women first movers.



Notes: *Error* takes integer values in $[-5, 5]$. Positive values indicate that the degree of risk aversion of the player was underestimated by their spouse, and negative values indicate overestimation. *Abs. error* takes integer values in $[0, 5]$ with higher values indicating greater deviation in predictions from spouse's actual choice.

[9]Appendix A shows the results of comparing the achieved risk exposure level with individual risk preferences. Similar deviations arise between the obtained household risk exposure and the first-mover's preferred risk exposure level.

The size of the first-mover's prediction error is dependent on the gender of the first-mover, the disparity in individual risk preferences between spouses, and the strategy of the second-mover. Table 3 lists the associations between the size of the *Error* and *Abs. Error* and the determinants. In column (5), one standard deviation increase in the disparity between the risk aversion of the second- and first-mover is associated with a 0.07 standard deviation increase in *Abs. error*, which measures how much the second-mover's lottery choice deviates from the first-mover's expectation. The results in columns (2)-(3) imply that second-movers who are less risk-averse than their spouse make lottery choices that are more risky than expected by their spouse and vice versa. The gender of the first-mover (and second-mover) is also an important determinant of prediction errors. Column (5) shows that when women are first-movers, the absolute difference between the second-mover's chosen lottery number and the expected lottery number is higher by 0.12 standard deviations. However, the gendered effects disappear after controlling for the second-mover's strategy. For every standard deviation increase in the *Strategy* variable causes the *Abs. error* to decline by 0.21 standard deviations. Between preference disparity and the second-mover's strategy, the strategy has a much larger impact on whether the first-mover expects the second-mover's choice.

Table 3: Determinants of prediction error.

	(1)	(2)	(3)	(4)	(5)	(6)
	Error	Error	Error	Abs. error	Abs. error	Abs. error
Woman first-mover (0/1)	0.029	0.032	0.037	0.123***	0.123***	0.068
	(0.048)	(0.047)	(0.047)	(0.047)	(0.047)	(0.047)
Pref. disparity		0.180***	0.179***			
		(0.023)	(0.023)			
Abs. pref. disparity					0.071***	0.059**
					(0.024)	(0.023)
Spouse's *Strategy*			0.016			-0.213***
			(0.024)			(0.023)
Constant	-0.014	-0.016	-0.018	-0.062*	-0.062*	-0.034
	(0.034)	(0.033)	(0.033)	(0.034)	(0.033)	(0.033)
R-squared	0.000	0.033	0.033	0.004	0.009	0.053
Observations	1770	1770	1770	1770	1770	1770

Notes: The dependent variables and continuous independent variables are standardized to aid interpretation. Standard errors are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Column (1) of table 4 compares the average values of the *Strategy* variable between men and women second-movers. The average value of *Strategy* for women is 74% higher than for men, second-movers. Panels (a) and (c) of figure 5 are box and whiskers plots of the lottery numbers that men and women second-movers choose for every potential lottery choice of their first-mover. These are their strategies. Visually, the median lottery number when men are second-movers is far less responsive to the first-mover's choice than it is for women as second-movers. Women try to match their lottery choices to their husbands' choices. This could be due to greater confidence in spouse's judgment, or lower faith in their own financial competence, or differences in decision-making powers (Conlon et al., 2021; Abbink et al., 2020). However, the positive value of the variable *Strategy* for both genders indicates some level of trust among both genders in the first-mover's choice. Panels (b) and (d) show the distribution of expected lottery choices of men and women second-movers as elicited from their spouses. Comparing actual to expected choices reveals that women expect their spouse to align with their choice as the first-mover, but in reality, men's choices as the second-mover are not sensitive to the first-mover's choice (Panels (a) and (b)). Even though, as second-movers, men make choices that are independent of their wives' choices (Panel (a)), in their role as the first-mover, they correctly predict that their wives will align with their choices (Panel (d)). Thus, the gendered difference in how players react to their spouses' lottery choice causes the household portfolio to deviate from women's choices more than men's.

Table 4: *Strategy* variable: Actual and expected.

	Actual		Expected		(2) - (1)	
	Mean	Std. Dev.	Mean	Std. Dev.	Difference	Std. Err.
	(1)		(2)		(3)	
Men	0.184	(0.530)	0.355	(0.488)	0.171***	(0.023)
Women	0.321	(0.500)	0.267	(0.511)	-0.054**	(0.023)
Difference	-0.137***		0.088***			
Std. Err.	(0.024)		(0.024)			
Number of men	887		887			
Number of women	887		887			

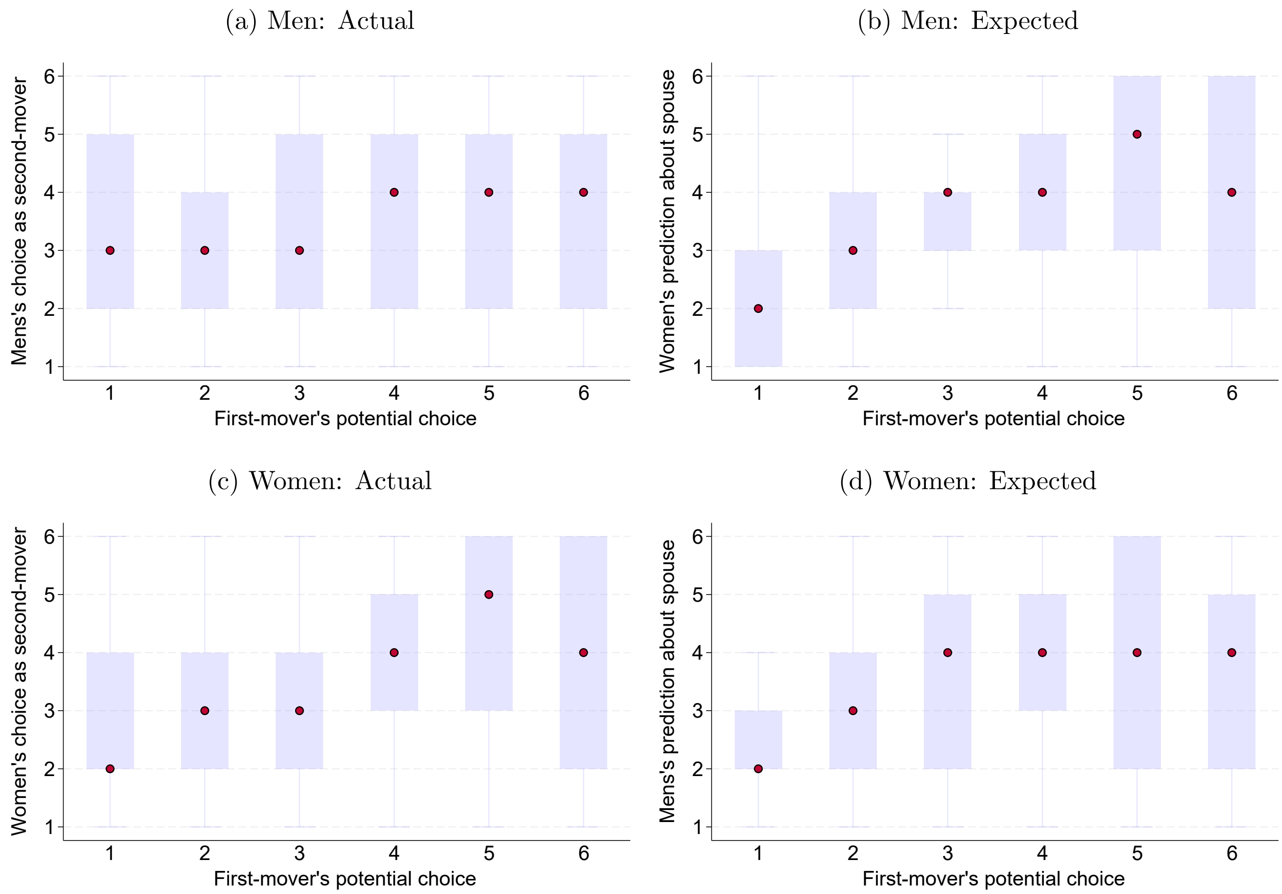
Notes: The *Strategy* variable takes values in $[-1, 1]$. Positive values indicate that the riskiness of second mover's choices co-move with the first mover's and negative values indicate that a second mover who tries to counter their spouse's choice. Standard errors and significance levels are based on t-tests. * p<0.1, ** p<0.05, and *** p<0.01.

4.1 Heterogeneity in second-mover's strategy

Table 5 reports associations between the second-mover's *Strategy* variable and the risk preference disparity with spouse, wife's intrahousehold decision-making powers, non-cognitive skills, and indicators of bargaining power.

Column (1) reports the associations from a pooled regression of the *Strategy* of men and women. The coefficient on the indicator for women second-movers is positive and statistically significant, indicating that women are more likely than men to align with the first-mover's choice. The decision-making power variable, which is constructed using women's answers to questions about the primary decision-maker for household decisions, is not statistically significant. In observational studies by Jianakoplos and Bernasek (2008) and Guha Thakurta and Wesselbaum (2026b), intrahousehold decision-making powers also do not affect household portfolio compositions. However, even though the decision-making variable is not significant, the coefficient on the indicator for women is no longer significant after controlling for the second-mover and their spouse's employment status (Column (2)). The sub-sample

Figure 5: Actual and expected choices of second-movers in the *Joint* game.



Notes: Box and whisker plots of the stated or predicted strategies of second-movers. The median is represented by the red markers and outliers were excluded. The horizontal axis plots the potential lottery number of the first-mover. The vertical axis plots the chosen lottery number of the second-mover as stated by themselves (Panels (a) and (c)) or as predicted by their spouse (Panels (b) and (d)).

regressions for only women (Column (3)) and men second-movers (Column (4)) also confirm that in households where women work for a living, both women and men second-movers lottery choices are more independent of the first-mover's choice.

Among women, those with greater risk aversion show a greater tendency to counterbalance increasing riskiness in the first-mover's lottery choice (Column (3)). The whole-sample regressions also show that greater divergence in individual risk preferences reduces the tendency to align with the first mover's choice.

Non-cognitive skills are relatively enduring patterns of thoughts and behaviors that have been shown to affect labor market outcomes, health status, and even gender gaps in earnings (Vella, 2024; Becker et al., 2012; Flinn et al., 2018). They are also important in explaining how people coordinate with their spouse's lottery choices. *Conscientiousness*, which correlates with self-discipline, is associated with greater alignment with the first-mover's choices among women. In contrast *extraversion* among women is associated with greater

Table 5: Heterogeneity in second-mover's strategy.

	All		Women	Men
	(1)	(2)	(3)	(4)
Woman (0/1)	0.188**	0.182		
	(0.082)	(0.146)		
Abs. pref. disparity	-0.049**	-0.048**	-0.043	-0.055
	(0.023)	(0.024)	(0.032)	(0.034)
Ind. risk pref.	-0.081***	-0.080***	-0.132***	-0.022
	(0.024)	(0.024)	(0.032)	(0.035)
Woman's decision-powers	0.003	0.003	-0.010	0.020
	(0.024)	(0.024)	(0.033)	(0.035)
Non-cognitive skills:				
Conscientiousness	0.077***	0.079***	0.120***	0.032
	(0.027)	(0.027)	(0.037)	(0.041)
Extraversion	-0.058**	-0.058**	-0.094***	0.006
	(0.024)	(0.024)	(0.032)	(0.039)
Agreeableness	0.008	0.008	0.014	-0.005
	(0.026)	(0.026)	(0.037)	(0.038)
Openness	0.047*	0.048*	-0.017	0.124***
	(0.026)	(0.026)	(0.035)	(0.038)
Neuroticism	-0.010	-0.009	0.006	-0.008
	(0.025)	(0.025)	(0.034)	(0.038)
Own occupation (Ref. Self-employed):				
Wage labor/Raise livestock (0/1)		-0.037	-0.351**	0.101
		(0.099)	(0.176)	(0.127)
At home (0/1)		0.041	-0.178	0.295
		(0.104)	(0.146)	(0.253)
Spouse occupation (Ref. Self-employed):				
Wage labor/Raise livestock (0/1)		-0.071	0.153	-0.468**
		(0.099)	(0.118)	(0.188)
At home (0/1)		0.033	0.028	-0.215
		(0.104)	(0.236)	(0.155)
Constant	-0.168*	-0.173	0.075	0.142
	(0.087)	(0.129)	(0.185)	(0.197)
R-squared	0.048	0.049	0.062	0.045
Observations	1770	1770	885	885

Notes: The dependent variables and continuous independent variables are standardized to aid interpretation. Controls for age difference between spouses, wealth status, household composition and education of subject and spouse were included in each specification. Standard errors are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

independence in choices as the second-mover. Among men, *openness to new experiences* is positively associated with a higher tendency to align with their wife's choice. These effects are robust to controls for intrahousehold decision-powers, household wealth status, and risk preferences.

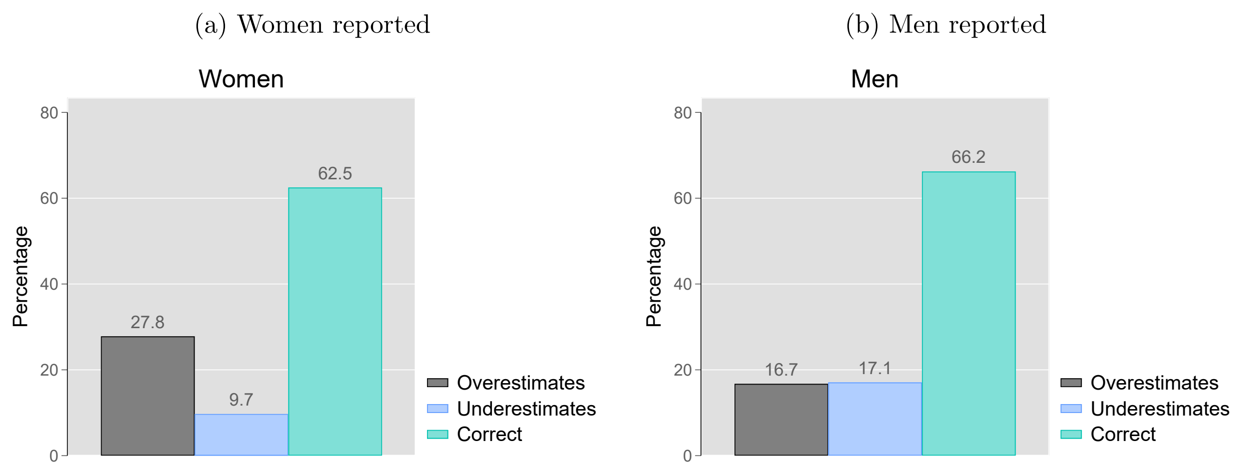
5 Risk-taking decisions in Sample Households

The experiment models household interactions in which partners' risk-taking decisions affect both partners' welfare, and at least one member's actions are not directly observable. To assess the external validity of the experimental findings, data on household assets and liabilities, as well as behavior under exogenous stress, were collected. Figures 6a and 6b report the percentage of men and women who correctly reported if their spouse made any savings in the past 12 months, and those who were wrong. 38% of women and 34% of men were mistaken about whether their spouse made any savings during the past twelve months. Women, in particular, were more likely to overestimate than underestimate whether their spouse had made savings. Women also underreported the number of livestock the household owned, including poultry, cows, and goats. Livestock assets are held as a source of informal savings and insurance (Pica-Ciamarra et al., 2011). Of the 604 couples in which at least one member reported taking a loan in the past twelve months, 60% disagreed on the number of loans taken out or still outstanding.

Figures 7a and 7c plot the distributions of women's and men's self-reported reasons for maintaining savings. Figures 7b and 7d plot the distributions of reasons for women and men maintaining savings as reported by their spouses. Men overestimate their wives' savings for farm investments by almost 20 percentage points, and underestimate their wives' savings for household emergencies and children's education-related expenses by 14% and 17%, respectively. Women overestimate how much their husbands save for household emergencies and their children's education. Table 6 reports associations between players' prediction errors in the game and the likelihood of mistaking their partner's saving behavior in the household. The size of men's prediction errors is positively associated with the likelihood of being mistaken about their wives' saving for children's education and marriage, which are things in which a significant fraction of women are solely responsible for deciding about (Figure 1). Whereas women's prediction errors are associated with a greater number of saving decisions, including investments, emergency funds, and loan repayments. This is because in the social context in Bangladesh, men generally have greater decision-making powers, and hence there is a greater opportunity of women being misinformed about a larger number of household affairs.

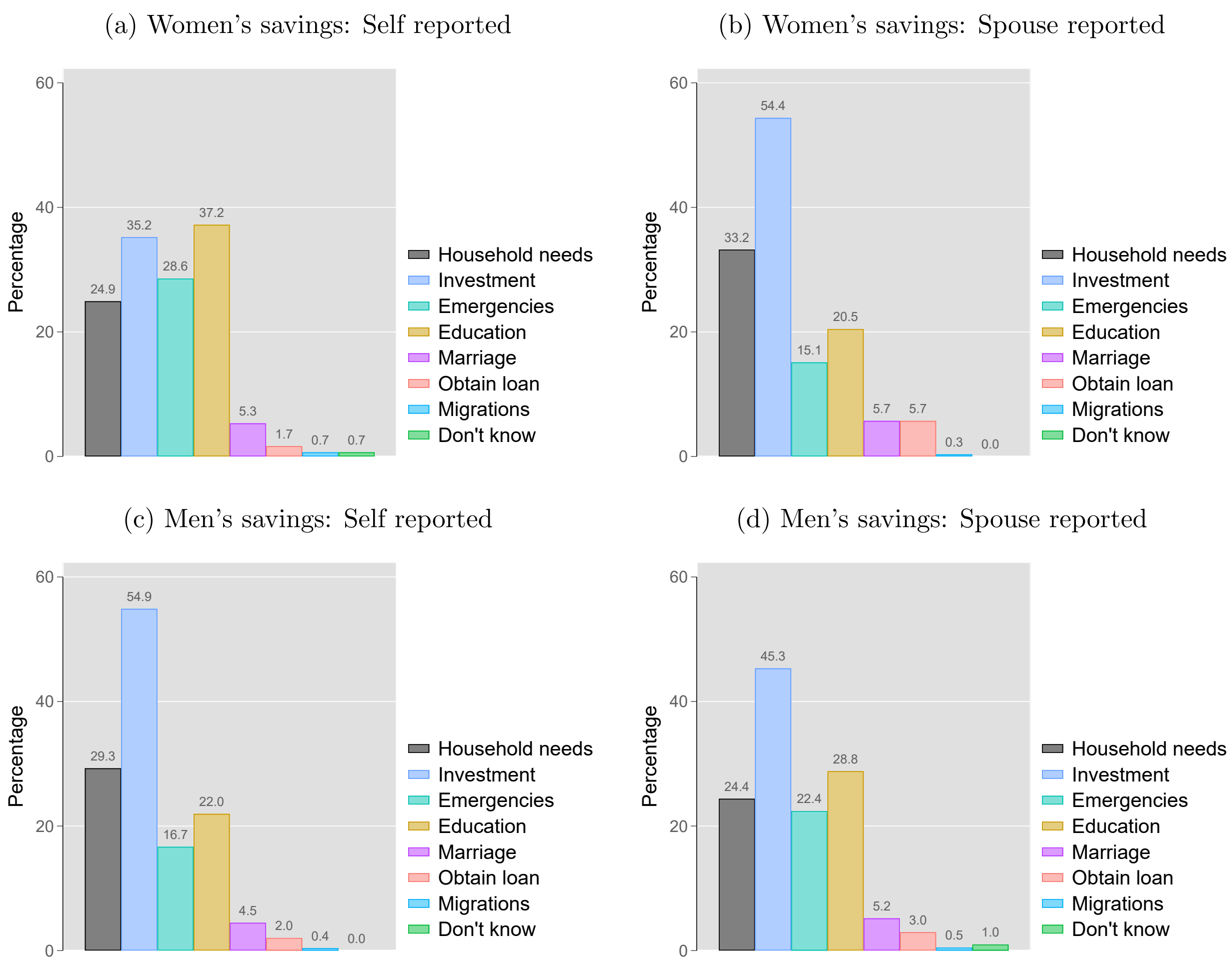
Appendix figure B.2 shows that a similar level of asymmetry in information between partners emerges when comparing people's self-reported and spouse-reported reasons for taking loans.

Figure 6: Reported savings behavior of men and women.



Notes: Husbands and wives were asked if either "you or your husband made any savings in the past 12 months?" If a person reported their spouse as having savings when their spouse reported not it was taken to be an overestimation, and if a person reported their spouse as not having savings when their spouse reported having savings it was taken to be an overestimation.

Figure 7: Reported purpose for maintaining savings of men and women.



Notes: If a person reported either themselves or their spouse as having made savings in the past 12 months, they were asked the top two reasons for maintaining the savings.

Table 6: Results from regression of mistaken beliefs about partner's savings for various purposes in the past twelve months on prediction errors (*Abs. error*).

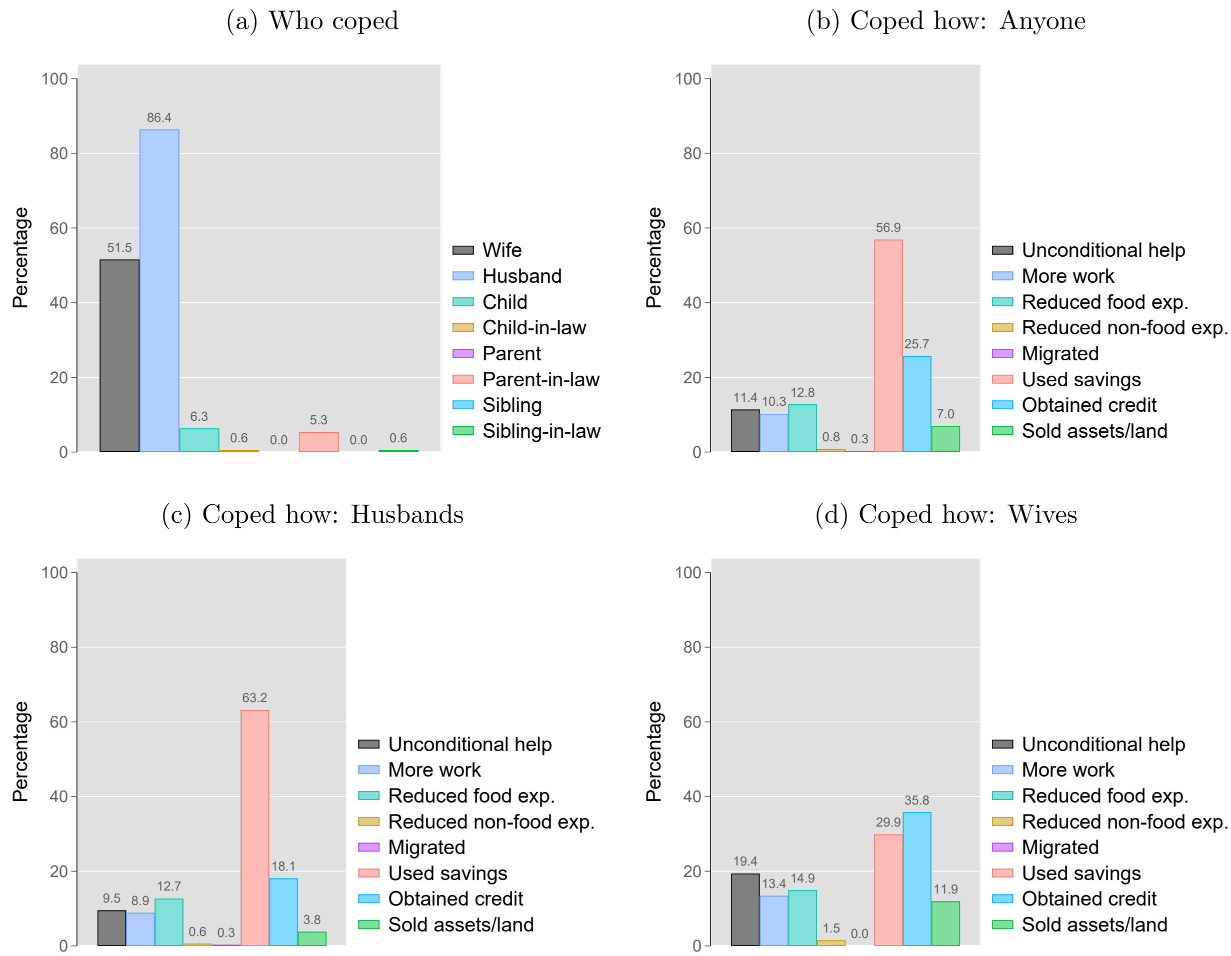
	Panel A: *Men's predictions about wife's savings*						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Household	Investment	Emergencies	Education	Marriage	Loan	Migration
Abs. error	0.014	-0.014	-0.008	-0.040**	-0.031*	-0.002	-0.005
	(0.020)	(0.020)	(0.019)	(0.017)	(0.017)	(0.018)	(0.015)
Constant	0.205	0.220	0.245	0.114	0.264	0.117	0.261*
	(0.187)	(0.191)	(0.165)	(0.163)	(0.161)	(0.168)	(0.155)
R-squared	0.136	0.077	0.298	0.309	0.335	0.339	0.422
Observations	678	678	678	678	678	678	678
	Panel B: *Women's predictions about husband's savings*						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Household	Investment	Emergencies	Education	Marriage	Loan	Migration
Abs. error	0.006	-0.042**	-0.041***	-0.003	-0.022	-0.029**	-0.025**
	(0.017)	(0.018)	(0.016)	(0.017)	(0.013)	(0.012)	(0.011)
Constant	0.277	0.370**	0.438***	0.337*	0.349***	0.331**	0.398***
	(0.177)	(0.183)	(0.161)	(0.172)	(0.134)	(0.132)	(0.113)
R-squared	0.238	0.195	0.341	0.278	0.500	0.554	0.591
Observations	678	678	678	678	678	678	678

Notes: The dependent variables are indicator variables for whether a husband is able to correctly predict the reason for their wife maintaining savings. Column 1 to 7 respectively are for savings to meet household needs, for agricultural or non-agricultural investments, for emergencies, for children's education, for their marriage, for loans and for migration. *Abs. error* is standardized to aid interpretation of estimates. Controls include individual risk preferences, age, education, and occupation of player and spouse, household composition, decision-making powers of wife and wealth index. Sample excludes couples who both stated that no one in the household made any savings in the past twelve months. Robust standard errors reported in parentheses. * $p<0.1$, ** $p<0.05$, and *** $p<0.01$.

5.1 How does failure to coordinate affect households?

As second-mover strategies show, people adapt their risk-taking decisions to changing information about their partner's risk-taking. As first movers, people making decisions under a false impression about their spouse's risk exposure can further increase household risk exposure. The results of this can manifest during periods of stress, such as weather shocks, a sudden illness in a family member, or job loss. The most common shocks faced in the past 12 months were weather events (26.4% of households), serious illness or death of an earning member (17.3%), serious illness or death of a non-earning member (11.0%), and the July-August revolution (10.7%). Panel (a) of figure 8 shows that partners of both genders played a role in taking coping measures following adverse shocks. Using savings was the preferred method used by men, while women, in addition to using available savings, also prefer getting credit (Panels (c) and (d)).

Figure 8: Shocks to the household: who coped and how.



Notes: Women were asked to report if in the past 12 months the household had been adversely affected any of the following shocks: weather events, natural disasters, serious illness or death of an earning member, serious illness or death of other, or the July-August political revolution. Panel a shows the distribution of household members who primarily helped cope with the shock across all shocks. Panel b shows the distribution of the measures used to cope across all shocks. Panel c shows the same distribution but only for the shocks where the husband was one of the members who primarily helped cope but not the wife. 315 out of 712 shocks reported fit this description. Panel d shows the the same distribution but for the shocks where the wife was one of the members who primarily helped cope but not the husband. 67 out of 712 reported shocks that fit this description. A household may have multiple shocks during the past 12 months.

Table 7 reports the associations between the absolute value of prediction errors (*Abs. error*) and the type of coping measure used. Couples who hold more biased beliefs about their spouse's actions in the game are less likely to rely on savings and more likely to have to sell household assets or obtain loans or even migrate following an adverse event. For a unit standard deviation increase in the size of the bias in women's *Abs. error* the likelihood of the household having to sell assets following an adverse event increases by 2.5 percentage points, and the likelihood of migration increases by 0.3 percentage points. For men, a standard deviation increase in *Abs. error* is associated with a 0.4 percentage point increase in the likelihood of migration.

Thus, couples who are worse at coordinating risk-taking decisions and therefore sharing risk between them are likelier to face insolvency in times of crisis, which in the extreme may even lead to migrating from their current place of residence. These figures likely underestimate the extent of the issue, as they are based solely on the coping measures reported by women. They do not account for hidden action by husbands.

Households with more risk-averse women are also more likely to rely on credit and less likely to use savings to tide themselves over. The majority of loans made out by microcredit organizations in Bangladesh are targeted at women, whereas men are more likely to have access to ready cash. More cautious women may appear more trustworthy to lenders and be more likely to obtain a loan in times of distress.

Table 7: Results from regression of type of coping measures used as reported by women on the absolute value of mistaken beliefs and individual risk preferences.

	(1) Help	(2) Get work	(3) Reduce food	(4) Reduce non-food	(5) Migrate	(6) Use savings	(7) Credit	(8) Sell assets
Wife's *Abs. error*	0.949	0.879	-1.191	0.155	0.327**	1.057	-2.413	2.514**
	(1.334)	(1.270)	(1.257)	(0.447)	(0.155)	(2.220)	(1.975)	(1.148)
Husband's *Abs. error*	0.656	0.156	-0.011	-0.061	0.410***	-6.495***	3.589*	0.276
	(1.234)	(1.104)	(1.626)	(0.479)	(0.154)	(2.163)	(1.915)	(1.149)
Wife's risk pref.	-1.218	2.003	-1.576	-0.385	-0.083	-4.750**	5.344**	-0.933
	(1.362)	(1.359)	(1.370)	(0.295)	(0.160)	(2.319)	(2.300)	(1.099)
Husband's risk pref.	-1.537	-0.770	0.789	0.669	0.276*	0.037	-1.261	0.833
	(1.300)	(1.263)	(1.436)	(0.422)	(0.154)	(2.179)	(1.960)	(0.988)
Shock type (Ref.: July-August '24[a]):								
Weather shock	-1.876	2.491	-12.606***	-0.872	-1.302***	4.174	-7.394	0.740
	(3.271)	(3.683)	(4.708)	(1.745)	(0.481)	(5.629)	(5.127)	(2.919)
Natural disaster	-1.301	-4.257	-18.415***	-2.064	-1.181*	3.262	2.139	-0.539
	(4.078)	(3.900)	(5.588)	(1.800)	(0.622)	(7.508)	(6.982)	(4.159)
Job loss	9.657	17.957***	-8.290	-2.494	-0.998	-13.900*	-4.829	-2.843
	(5.932)	(6.081)	(6.483)	(1.619)	(0.657)	(7.615)	(7.359)	(4.137)
Illness of non-earner	12.469***	-6.756*	-20.556***	-0.498	-1.231**	6.057	-2.223	5.816*
	(4.352)	(3.478)	(4.872)	(2.263)	(0.526)	(6.328)	(6.093)	(3.496)
Illness of earner	16.306***	-4.738	-23.988***	-1.753	-1.220**	3.895	-7.625	2.882
	(5.357)	(3.487)	(5.481)	(1.536)	(0.579)	(7.075)	(6.630)	(4.174)
Constant	28.547**	13.501	24.187*	1.024	-1.248	51.594***	54.897***	-2.793
	(11.088)	(11.988)	(14.111)	(4.616)	(1.394)	(19.101)	(14.839)	(7.394)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Village FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.142	0.178	0.174	0.051	0.073	0.190	0.142	0.124
Observations	709	709	709	709	709	709	709	709

Notes: Estimated coefficients denote change in probability of an event in percentage points. *Abs. error* and γ variables are standardized to aid interpretation of estimates.Controls include age, education, and occupation of player and spouse, household composition and wealth index. SEs reported in parentheses. * p<0.1, ** p<0.05, and *** p<0.01.

Notes: Estimated coefficients denote change in probability of an event in percentage points. *Abs. error* and γ variables are standardized to aid interpretation of estimates. SEs reported in parentheses. * p<0.1, ** p<0.05, and *** p<0.01.

[a] The reference category for the shocks is the political revolution and associated socioeconomic upheaval during July-August 2024.

6 Conclusions

This paper uses a lab-in-field experiment to analyze coordination in risk-taking decisions between spouses in rural Bangladesh. I find evidence of widespread coordination error between spouses. Using a unique sequential lottery game, I directly observe how spouses respond to each other in joint risk-taking decisions. The findings reveal that men are less willing than women to align with their spouse's decisions. Consequently women face greater difficulty in coordinating with their spouse's choice and household risk exposure levels deviate further from the first-mover's preferred risk level when women perform this role. These findings contribute to the growing literature on intra-household information frictions (Conlon et al., 2021; Tagat et al., 2024; Abbink et al., 2020) by demonstrating how, under experimental conditions, frictions in information pass-through between spouses can contribute to coordination errors in risk-taking decisions in households.

Household data confirms the gendered division of household tasks, with most outdoor activities being performed solely by men, and activities such as investing in education and health of children often being performed by women alone. I show that the division of roles in the household fosters asymmetric information between spouses, especially with regard to the state of household assets–both pecuniary savings and real assets such as livestock. I am able to relate the experimental results to real-world data, showing that women who overestimated their husband's risk aversion in the game were also more likely to be overoptimistic about their spouse's savings strategy in the household. As women are more likely to make decisions such as taking out loans for their children's future and make provisions for possible emergencies, such mistaken beliefs can threaten to overexpose the household to risk, especially in times of preexisting stress due to adverse shocks such as droughts, which are common among agricultural households.

While the results derived in this study are based on a sample of participants recruited from rural Bangladesh, its findings are relevant to household decision-making in all information-constrained environments. Especially, patriarchal norms by restricting women's roles in various spheres of decision-making exacerbate the information dearth for them and make it harder for them to make well-informed decisions. The present study shows how, in addition to limiting women's agency, gender norms also negatively impact household well-being.

References

Abbink, K., Islam, A., and Nguyen, C. (2020). Whose voice matters? An experimental examination of gender bias in intra-household decision-making. *Journal of Economic Behavior & Organization*, 176:337–352.

Anderson, S., Beaman, L. G., and Platteau, J.-P., editors (2018). *Towards gender equity in development*. UNU-WIDER studies in development economics. Oxford University Press, Oxford, United Kingdom, first edition edition.

Bangladesh Bureau of Statistics (2024). Population and Housing Census 2022: District Report: Bogura. Technical report, Bangladesh Bureau of Statistics.

Becker, A., Deckers, T., Dohmen, T., Falk, A., and Kosse, F. (2012). The Relationship Between Economic Preferences and Psychological Personality Measures. *Annual Review of Economics*, 4(Volume 4, 2012):453–478.

Becker, G. (1981). *A treatise on the family*, volume 30. Harvard University Press. Publication Title: Cambridge, MA.

Binswanger, H. P. (1981). Attitudes Toward Risk: Theoretical Implications of an Experiment in Rural India. *The Economic Journal*, 91(364):867.

Browning, M. and Chiappori, P. A. (1998). Efficient Intra-Household Allocations: A General Characterization and Empirical Tests. *Econometrica*, 66(6):1241.

Chakravarty, S., Harrison, G. W., Haruvy, E. E., and Rutström, E. E. (2011). Are You Risk Averse over Other People's Money? *Southern Economic Journal*, 77(4):901–913.

Charness, G., Gneezy, U., and Kuhn, M. A. (2012). Experimental methods: Between-subject and within-subject design. *Journal of Economic Behavior & Organization*, 81(1):1–8.

Chiappori, P.-A. (1992). Collective Labor Supply and Welfare. *Journal of Political Economy*.

Chiappori, P.-A. (1997). Collective models of household behavior: the sharing rule approach. *Intrahousehold resource allocation in developing countries*, pages 39–52.

Choudhury, M. S. (2019). MFI microsavings: inclusive or transformative? – A participatory survey of household savings in northeast rural Bangladesh. 1(1).

Chowdhury, S., Sutter, M., and Zimmermann, K. F. (2022). Economic Preferences across Generations and Family Clusters: A Large-Scale Experiment in a Developing Country. *Journal of Political Economy*, 130(9):2361–2410.

Coller, M. and Williams, M. B. (1999). Eliciting Individual Discount Rates. *Experimental Economics*, 2(2):107–127.

Conlon, J., Mani, M., Rao, G., Ridley, M., and Schilbach, F. (2021). Learning in the Household. Technical Report w28844, National Bureau of Economic Research, Cambridge, MA.

Flinn, C. J., Todd, P. E., and Zhang, W. (2018). Personality traits, intra-household allocation and the gender wage gap. *European Economic Review*, 109:191–220.

Guha Thakurta, A. and Wesselbaum, D. (2026a). Portfolio choice with intra-household bargaining and gender differences in preferences. *Scottish journal of political economy*.

Guha Thakurta, A. and Wesselbaum, D. (2026b). Portfolio Choice With Intra-Household Bargaining and Gender Differences in Preferences. *Scottish Journal of Political Economy*, n/a(n/a):e70047.

Harrison, G. W., Lau, M. I., Rutström, E. E., and Sullivan, M. B. (2005). Eliciting risk and time preferences using field experiments: Some methodological issues. In *Field Experiments in Economics*, volume 10, pages 125–218. Emerald Group Publishing Limited.

Harrison, G. W. and Rutström, E. E. (2008). Risk Aversion in the Laboratory. In *Research in Experimental Economics*, volume 12, pages 41–196. Emerald (MCB UP), Bingley.

Holt, C. A. and Laury, S. K. (2002). Risk Aversion and Incentive Effects. *American Economic Review*, 92(5):1644–1655.

Holt, C. A. and Laury, S. K. (2014). Assessment and estimation of risk preferences. In Machina, M. J. and Viscusi, W. K., editors, *Handbook of the economics of risk and uncertainty*, volume 1, pages 135–201. North-Holland, Amsterdam, Netherlands.

Jianakoplos, N. A. and Bernasek, A. (2008). Family Financial Risk Taking When the Wife Earns More. *Journal of Family and Economic Issues*, 29(2):289–306.

Lundberg, S. and Pollak, R. A. (1993). Separate spheres bargaining and the marriage market. *Journal of Political Economy*, 101(6):988–1010.

Mazzocco, M. (2004). Saving, Risk Sharing, and Preferences for Risk. *American Economic Review*, 94(4):1169–1182.

Mazzocco, M. and Saini, S. (2012). Testing Efficient Risk Sharing with Heterogeneous Risk Preferences. *American Economic Review*, 102(1):428–468.

Mou, M. M., Islam, M. R., Mondal, T., Monalesa, N., and Abdullah, M. M. (2019). Improved Vegetable Cultivation Practices: An Adoption Study among the Farmers' of Some Selected Areas in Bangladesh. *Asian Journal of Agricultural Extension, Economics & Sociology*, 37(2):1–8.

Pica-Ciamarra, U., Tasciotti, L., Otte, J., and Zezza, A. (2011). Livestock Assets, Livestock Income and Rural Households: Cross-country Evidence from Household Surveys. ESA Working Paper 11-17, Food and Agriculture Organization of the United Nations (FAO), Rome.

Schaner, S. (2015). Do Opposites Detract? Intrahousehold Preference Heterogeneity and Inefficient Strategic Savings. *American Economic Journal: Applied Economics*, 7(2):135–174.

Tagat, A., Kapoor, H., and Kulkarni, S. (2024). Private lives: experimental evidence on information completeness in spousal preferences. *Journal of the Economic Science Association*, 10(1):62–75.

The Business Standard (2024). Divorce Rate Declined in 2023: BBS. *The Business Standard*.

Vella, M. (2024). The relationship between the Big Five personality traits and earnings: Evidence from a meta-analysis. *Bulletin of Economic Research*, 76(3):685–712. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/boer.12437.

Appendix A

A.1 Individual risk preferences

Players chose one out of a set of six risky lotteries (Table 2). Each lottery had two equally likely non-negative outcomes. Players were handed a sheet of paper with visual representations of the six options. Figure B.1 reproduces the sheet of paper handed to players with the visual representations. A lottery was depicted using a horizontal bar graph with an orange bar representing the payout in case of the "high" outcome and a white bar representing the payout in case of the "low" outcome. All six lotteries were represented in this fashion, so that individuals with limited literacy were not at a disadvantage. Enumerators also explained each of the lotteries while pointing them out in the sheet of paper. Any winnings from this game were equally divided between the player and their spouse. It is used to model individual decision-making for the household.

Based on the player's choice of lottery, the risk preference of the individual is calculated assuming constant relative risk aversion (CRRA) utility. CRRA utility functions are a special case of the harmonic absolute risk aversion (HARA) class of utility functions which also includes constant absolute risk aversion (CARA) functions. CRRA utility functions are commonly used to describe risk preferences derived in lab experiments (Chakravarty et al., 2011; Holt and Laury, 2002; Harrison et al., 2005; Coller and Williams, 1999; Holt and Laury, 2014; Harrison and Rutström, 2008).

$$U(x) = \frac{x^{\gamma}}{1-\gamma} \quad \text{where } \gamma \neq 1 \tag{2}$$

An individual's risk preference for the couple (or household) is measured based on their choice of lottery. I compare the expected utility from their revealed choice to the possible expected utilities from the other lotteries to find the values of γ consistent with their choice.

For example, if an individual chooses lottery 2 from Table 2, it means that for that individual:

$$\begin{aligned} EU(L_2) &\geq EU(L_1) \\ \implies 0.5 * \frac{190^{\gamma}}{1-\gamma} + 0.5 * \frac{100^{\gamma}}{1-\gamma} &\geq 0.5 * \frac{125^{\gamma}}{1-\gamma} + 0.5 * \frac{125^{\gamma}}{1-\gamma} \\ \implies \gamma &\leq 3.03 \end{aligned} \tag{3}$$

And,

$$
\begin{aligned}
EU(L_2) &\geq EU(L_3) \\
\implies 0.5 * \frac{190^{\gamma}}{1-\gamma} + 0.5 * \frac{100^{\gamma}}{1-\gamma} &\geq 0.5 * \frac{240^{\gamma}}{1-\gamma} + 0.5 * \frac{90^{\gamma}}{1-\gamma} \\
\implies \gamma &\geq 1.99
\end{aligned} \tag{4}
$$

Thus, their revealed risk aversion parameter (γ) value exists in the interval $[1.99, 3.03]$. For simplicity, a person's revealed risk preference level is characterized by the mid-point of the range of γ consistent with their observed choice. Higher values of the risk aversion parameter (γ) indicate greater aversion to risk. The parameter calculated based on a player's choice in this game is denoted as γ_{Ind} since it represents individual risk preferences.

In case a person chooses either lottery number 1 or 6, it isn't possible to calculate a mid-point value of γ, so either the upper- or lower-limit of the relevant interval of values is used, based on whichever is finite. This design upper-codes γ values at 3.03 and lower-codes them at 0.12. To achieve greater precision in calculated γ values would have required a greater number of choices to be presented to players either in the same set or across sets. This was not possible due to constraints set by the field and the observed tendency of fatigue among players when faced with too many choices during piloting.

I have to make three assumptions to be able to characterize risk preferences based on the observed choices in this way:

1. Individual utilities satisfy constant relative risk aversion (CRRA).
2. Implicit in the above assumption is the condition that people are not altruistic about their spouse's income or risk preference.
3. Individuals do not make mistakes when choosing lotteries.

A.2 Household Risk Exposure Level

To measure the risk exposure level achieved by first-mover and second-movers together, I employ an intellectual abstraction. In the game, the first mover makes their choice from among the six options knowing that the second-mover will choose a lottery conditional on their choice as first-mover. The first-mover's choice will thus be influenced by their belief about their spouse's lottery choice. Incorrect beliefs can cause the first-mover to stray from their preferred risk level. I assume a central planner with perfect knowledge of the second-mover's strategy and calculate the risk parameter that would have led them to make the same lottery choice as the first-mover.

Say, that the second-mover's strategy is as follows:

- lottery 5 if the first-mover chooses 1,
- lottery 5 if the first-mover chooses 2,
- lottery 4 if the first-mover chooses 3,

- lottery 3 if the first-mover chooses 4,
- lottery 2 if the first-mover chooses 5,
- and lottery 1 if the first-mover chooses 6,

and the first-mover's choice is lottery 2. A central planner who knows the second-mover's strategy would know that choosing lottery 2 is tantamount to choosing the lottery combination (2, 5) out of the six alternative combinations created by the second-mover's strategy. Comparing the expected utilities calculated for each of the lottery combinations produces the range of γ values consistent with the central planner's choice. One of the comparisons for this case would be:

$$
\begin{aligned}
&EU(L_2, L_5) \geq EU(L_1, L_5)\\
&\implies 0.25 * \frac{(190+15)^\gamma}{1-\gamma} + 0.25 * \frac{(190+400)^\gamma}{1-\gamma} + 0.25 * \frac{(100+15)^\gamma}{1-\gamma} + 0.25 * \frac{(100+400)^\gamma}{1-\gamma}\\
&\quad \geq 0.25 * \frac{(125+0)^\gamma}{1-\gamma} + 0.25 * \frac{(125+450)^\gamma}{1-\gamma} + 0.25 * \frac{(125+0)^\gamma}{1-\gamma} + 0.25 * \frac{(125+450)^\gamma}{1-\gamma}\\
&\qquad \implies \gamma \in [0.01, 5.0]
\end{aligned} \tag{5}
$$

The intersection of the γ ranges obtained from pairwise comparisons of each expected combination is used to calculate the risk exposure levels achieved by the first- and second-movers. This parameter is denoted as γ_{Joint}. The values of γ consistent with each pairwise comparison of the entire set of possible combinations of lotteries in Table 2 were calculated using the `numpy` package in *Python*.

The obtained γ values are discretized into the intervals presented in Table 2. This is done to make the risk levels comparable to the individual risk preference values (γ_{Ind}).

A.3 Deviation from preferred risk exposure

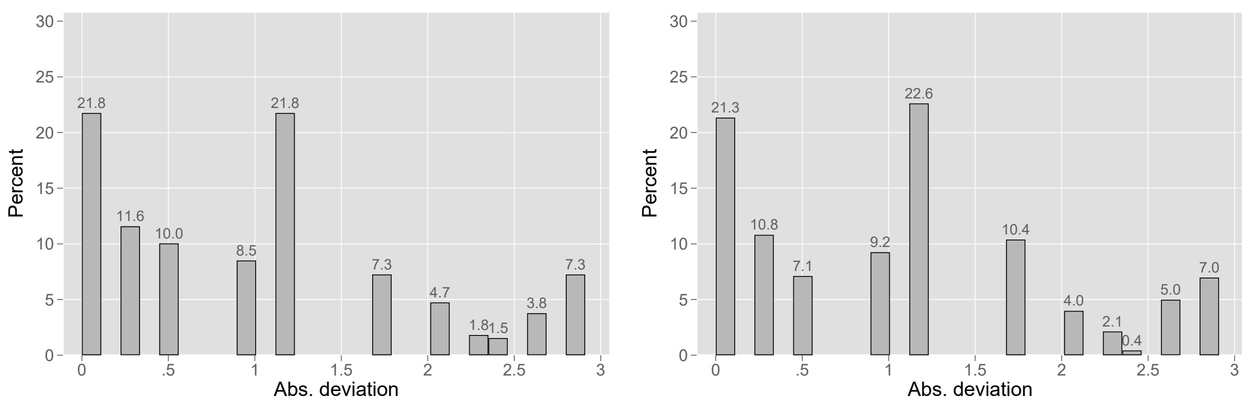
To estimate the effect of the coordination error on overall household risk exposure, I compare the individual risk aversion parameter (γ_{Ind}) of the first-mover with the risk aversion parameter that characterizes the final lottery combination that arises in the game. The absolute deviation in the risk exposure level achieved by the first-mover in the game from their revealed risk preference ($|\gamma_{Ind} - \gamma_{Joint}|$) captures how far the achieved exposure level was from their preferred level. Figure 9 presents the distribution of the absolute deviations of achieved level from preferred level for men and women first-movers. Only 22% of men first-movers (Sub-figure 9a) and 21% of women first-movers (Sub-figure 9b) are able to achieve their preferred risk exposure level[10].

[10] γ_{Joint} could be calculated for 717 (81%) out of the sample of 887 male first-movers and for 703 (79%) out of the sample of 887 female first-movers. This value could not be calculated if the choice of lottery combination in the *Joint* game did not follow CRRA utility function predictions.

Figure 9: Distribution of the absolute deviation of the risk exposure level achieved by first-movers in the game (γ_{Joint}) from their individual risk preference (γ_{Ind}).

(a) Men: $|\gamma_{Ind} - \gamma_{Joint}|$

(b) Women: $|\gamma_{Ind} - \gamma_{Joint}|$



Notes: γ_{Ind} represents individual risk preference levels of first-movers. γ_{Joint} represents the risk exposure level achieved by the first-mover and the second-mover together in the game. γ_{Joint} could be constructed from 717 (81%) of male first-movers and 703 (79%) of female first-movers.

Appendix B: Conceptual Framework

This section presents a simple conceptual framework to describe the decision-making process between couples in the experiment. I start with a basic model of sequential household decision making where:

- The **first-mover** (H) who has initial wealth M_H chooses what share (x_H) of it to allocate to a risky investment with returns: $X \sim f_X$.
- The **second-mover** (W) observes x_H and then chooses their investment allocation x_W into the risky investment.
- Each spouse **only maximizes their own expected utility** and this does not depend on the other's preferences.
- There is a sharing rule in the household such that the **second-mover shares** λ_W **share of their earnings with the spouse** and the **first-mover shares** λ_H **share of their earnings with the spouse.**

B.1 Earnings

Each spouse's earnings is a random variable given by sum of earnings from the investment and any money not invested:

First-mover:

$$y_H = x_H M_H X + (1 - x_H)M_H \tag{B.1}$$

Second-mover:

$$y_W = x_W M_W X + (1 - x_W)M_W \tag{B.2}$$

B.2 Consumption with Sharing Rule

Each spouse's consumption incorporates a fraction of the other's earnings:

First-mover:

$$c_H = (1 - \lambda_H)y_H + \lambda_W y_W \tag{B.3}$$

Second-mover:

$$c_W = (1 - \lambda_W)y_W + \lambda_H y_H \tag{B.4}$$

B.3 Preferences

Each individual i maximizes a CRRA utility which is a function of their own consumption:

$$U_i(c) = \frac{c_i^{1-\rho_i}}{1 - \rho_i}, \text{ for } i = H, W \tag{B.5}$$

where $\rho_i \neq 1$.

B.4 Second Mover's Optimization Problem

The second-mover maximizes:

$$\max_{x_W \in [0,1]} \quad E\left[U_W(y_W, y_H)\right] = E\left[\frac{((1-\lambda_W)y_W + \lambda_H y_H)^{1-\rho_W}}{1-\rho_W}\right]. \tag{B.6}$$

The first-order condition is[11]:

$$\begin{aligned}
\frac{\partial}{\partial x_W} E\left[\frac{c_W^{1-\rho_W}}{1-\rho_W}\right] &= 0 \\
\implies E\left[(c_W)^{-\rho_W}(1-\lambda_W)M_W(X-1)\right] &= 0 \\
\implies (1-\lambda_W)M_W E\left[(c_W)^{-\rho_W}(X-1)\right] &= 0 \\
\implies E\left[(c_W)^{-\rho_W}(X-1)\right] &= 0 \qquad \text{(B.7)}
\end{aligned}$$

This defines $x_W^*(x_H)$, the second mover's best response and so also $y_W^*(x_W^*(x_H))$.

$$y_W^*(x_H) = x_W^*(x_H)M_W X + (1 - x_W^*(x_H))M_W \tag{B.8}$$

$$\implies \frac{dy_W^*(x_H)}{dx_H} = M_W(X-1)\frac{dx_W^*(x_H)}{dx_H} \tag{B.9}$$

B.4.1 Implicit Function Theorem to calculate $\frac{dx_W^*}{dx_H}$:

Rewriting F.O.C. in equation (B.7) as:

$$F(x_W, x_H) = (1-\lambda_W)M_W E\left[(c_W)^{-\rho_W}(X-1)\right] = 0 \tag{B.10}$$

Using IFT:

$$\frac{dx_W^*}{dx_H} = -\frac{\frac{\partial F}{\partial x_H}}{\frac{\partial F}{\partial x_W}} \tag{B.11}$$

$$\tag{B.12}$$

$$\frac{\partial F}{\partial x_H} = -(1-\lambda_W)M_W E[\rho_W (c_W)^{-\rho_W - 1}(X-1)\frac{\partial c_W}{\partial x_H}] \tag{B.13}$$

[11]In order to exchange the derivative and expectations operators, it is necessary that the conditions for the Dominated Convergence Rule hold (?). I assume that this condition is satisfied for each function within the expectation operator in this section.

$$\frac{\partial F}{\partial x_W} = -(1-\lambda_W)M_W E[\rho_W (c_W)^{-\rho_W - 1}(X-1)\frac{\partial c_W}{\partial x_W}] \tag{B.14}$$

Where,

$$\frac{\partial c_W}{\partial x_H} = \frac{\partial}{\partial x_H}[(1-\lambda_W)y_W + \lambda_H y_H] = \lambda_H M_H (X-1) \tag{B.15}$$

$$\frac{\partial c_W}{\partial x_W} = \frac{\partial}{\partial x_W}[(1-\lambda_W)y_W + \lambda_H y_H] = (1-\lambda_W) M_W (X-1) \tag{B.16}$$

Substituting equation (B.15) in (B.13) and equation (B.16) in (B.14),

$$\frac{\partial F}{\partial x_H} = -(1-\lambda_W)\lambda_H M_W M_H \rho_W E[(c_W)^{-\rho_W - 1}(X-1)^2] \tag{B.17}$$

$$\frac{\partial F}{\partial x_W} = -(1-\lambda_W)^2 M_W^2 \rho_W E[(c_W)^{-\rho_W - 1}(X-1)^2] \tag{B.18}$$

From equations (B.11), (B.17), and (B.18),

$$\frac{dx_W^*}{dx_H} = -\frac{-(1-\lambda_W)\lambda_H M_W M_H \rho_W E[(c_W)^{-\rho_W - 1}(X-1)^2]}{-(1-\lambda_W)^2 M_W^2 \rho_W E[(c_W)^{-\rho_W - 1}(X-1)^2]}$$

Simplifying,

$$\boxed{\frac{dx_W^*}{dx_H} = -\frac{\lambda_H M_H}{(1-\lambda_W)M_W}} \tag{B.19}$$

Given $\lambda_W < 1$, $\frac{dx_W^*}{dx_H} < 0$, implying that the second mover reduces the share of own income invested in the risky asset in response to higher investment by the first mover. Further, the derivative is equal to the ratio between the amount of earnings shared by the first-mover and the amount of earnings retained by the second-mover. Thus, for couples which share a larger fraction of earnings with each other the responsiveness of the second-mover's strategy to the first-mover's choice is greater.

So, from equations (B.43) and (B.19) we have:

$$\begin{aligned}\frac{dy_W^*(x_H)}{dx_H} &= M_W(X-1)\frac{dx_W^*(x_H)}{dx_H}\\ &= -\frac{\lambda_H M_H(X-1)}{(1-\lambda_W)}\end{aligned} \tag{B.20}$$

Implicit Function Theorem to calculate $\frac{dx_W^*}{d\rho_W}$ Using IFT on second mover's F.O.C. in equation (B.44):

$$\frac{dx_W^*}{d\rho_W} = -\frac{\frac{\partial F}{\partial \rho_W}}{\frac{\partial F}{\partial x_W}} \tag{B.21}$$

$$\begin{aligned}\frac{\partial F}{\partial \rho_W} &= \frac{\partial}{\partial \rho_W}\left[(1-\lambda_W)M_W E\left\{(c_W)^{-\rho_W}(X-1)\right\}\right]\\ &= -(1-\lambda_W)M_W E[\ln c_W (c_W)^{-\rho_W}(X-1)]\end{aligned} \tag{B.22}$$

Thus, using the result from equation (B.18):

$$\frac{dx_W^*}{d\rho_W} = -\frac{-(1-\lambda_W)M_W E[(X-1)(c_W)^{-\rho_W}\ln c_W]}{-(1-\lambda_W)^2 M_W^2 \rho_W E[(c_W)^{-\rho_W-1}(X-1)^2]}$$

Simplifying,

$$\boxed{\frac{dx_W^*}{d\rho_W} = -\frac{1}{(1-\lambda_W)M_W\rho_W}\cdot\frac{E\left[(X-1)(c_W)^{-\rho_W}\ln c_W\right]}{E\left[(X-1)^2(c_W)^{-\rho_W-1}\right]}} \tag{B.23}$$

The second term on the RHS of equation (B.23) is a ratio between a term which captures how the marginal utility of the second-mover varies with respect to their degree of risk aversion and a term which is a kind of variance weighted utility measure. The first term is the inverse product of the share of own income that the second-mover retains ($(1-\lambda_W)M_W$) and the second-mover's risk aversion parameter. In the second term, the denominator which is a product of a squared term and a transformation of c_w (> 0) is non-negative. The numerator's is given by:

$$E\left[(X-1)(c_W)^{-\rho_W}\ln c_W\right] = -\frac{1}{(1-\lambda_W)M_W}\cdot\frac{\partial^2 E\left[U_W(y_W,y_H)\right]}{\partial\rho_W\partial x_W}$$

where $\frac{\partial^2 E[U_W(y_W,y_H)]}{\partial\rho_W\partial x_W}$ can be interpreted as the change in the marginal expected utility of the share of wealth in the risky asset due to an increase in the second mover's level of risk aversion. It stands to reason then that $\frac{\partial^2 E[U_W(y_W,y_H)]}{\partial\rho_W \partial x_W} < 0$. Therefore, the numerator, $E\left[(X-1)(c_W)^{-\rho_W}\ln c_W\right]$, is also non-negative in sign.

B.5 First Mover's Optimization Problem

The first mover anticipates $x_W^*(x_H)$ and maximizes:

$$\max_{x_H \in [0,1]} \quad E\left[U_H(y_W, y_H)\right] = E\left[\frac{((1-\lambda_H)y_H + \lambda_W y_W^*(x_H))^{1-\rho_H}}{1-\rho_H}\right]. \tag{B.24}$$

The first-order condition is:

$$(1-\lambda_H)M_H E\left[(c_H)^{-\rho_H}(X-1)\right] + \lambda_W E\left[(c_H)^{-\rho_H}\frac{dy_W^*}{dx_H}\right] = 0 \tag{B.25}$$

Substituting $\frac{dy_W^*(x_H)}{dx_H}$ from equation (B.20) in (B.25):

$$\begin{aligned}
(1-\lambda_H)M_H E\left[(c_H)^{-\rho_H}(X-1)\right] + \lambda_W E\left[(c_H)^{-\rho_H}\left\{-\frac{\lambda_H M_H (X-1)}{(1-\lambda_W)}\right\}\right] &= 0 \\
\implies (1-\lambda_H)M_H E\left[(c_H)^{-\rho_H}(X-1)\right] - \frac{\lambda_W\lambda_H M_H}{1-\lambda_W}E\left[(c_H)^{-\rho_H}(X-1)\right] &= 0 \\
\implies \left[(1-\lambda_H)M_H - \frac{\lambda_W\lambda_H M_H}{1-\lambda_W}\right]E\left[(c_H)^{-\rho_H}(X-1)\right] &= 0 \\
\implies \frac{(1-\lambda_W-\lambda_H)M_H}{1-\lambda_W}E\left[(c_H)^{-\rho_H}(X-1)\right] &= 0 \qquad \text{(B.26)}
\end{aligned}$$

If $\lambda_H + \lambda_W = 1$ the F.O.C. (B.26) holds for all values of c_H and x_H. The condition $\lambda_H + \lambda_W = 1$ implies that the share of own income that individual i retains (λ_i) for themselves is equal to the share of the spouse j's income that is transferred to them $(1-\lambda_j)$ and vice versa. I make the assumption that $\lambda_H + \lambda_W \neq 1$ since it is necessary to derive analytical solutions to the comparative static conditions. Further, if $\lambda_H + \lambda_W > 1$, then at least one of λ_H and λ_W must be ≥ 0.5. I do not expect sharing between spouses to be that large in the present context, so I assume that $\lambda_H + \lambda_W < 1$.

B.5.1 Implicit Function Theorem to calculate $\frac{dx_H^*}{d\rho_W}$

Rewriting FOC as:

$$G(x_H, \rho_W) = \frac{(1-\lambda_W-\lambda_H)M_H}{1-\lambda_W}E\left[(c_H(x_H,\rho_W))^{-\rho_H}(X-1)\right] = 0. \tag{B.27}$$

Using IFT:

$$\frac{dx_H^*}{d\rho_W} = -\frac{\frac{\partial G}{\partial \rho_W}}{\frac{\partial G}{\partial x_H}} \tag{B.28}$$

$$\frac{\partial G}{\partial \rho_W} = \frac{(1-\lambda_W-\lambda_H)M_H}{1-\lambda_W} E\left[(X-1)(-\rho_H)(c_H)^{-\rho_H-1}\cdot\frac{\partial c_H}{\partial \rho_W}\right] \tag{B.29}$$

$$\frac{\partial G}{\partial x_H} = \frac{(1-\lambda_W-\lambda_H)M_H}{1-\lambda_W} E\left[(X-1)(-\rho_H)(c_H)^{-\rho_H-1}\cdot\frac{\partial c_H}{\partial x_H}\right] \tag{B.30}$$

Where,

$$\frac{\partial c_H}{\partial \rho_W} = \lambda_W\frac{dy_W^*}{d\rho_W} = \lambda_W M_W(X-1)\frac{dx_W^*}{d\rho_W} \tag{B.31}$$

$$\frac{\partial c_H}{\partial x_H} = (1-\lambda_H)\frac{\partial y_H}{\partial x_H} + \lambda_W\frac{dy_W^*}{dx_H} = (1-\lambda_H)M_H(X-1) + \lambda_W\frac{dy_W^*}{dx_H} \tag{B.32}$$

Substituting equation (B.23) in (B.31):

$$\begin{aligned}\frac{\partial c_H}{\partial \rho_W} &= \lambda_W M_W(X-1)\cdot\left[-\frac{1}{(1-\lambda_W)M_W\rho_W}\cdot\frac{E\left\{(X-1)(c_W)^{-\rho_W}\ln c_W\right\}}{E\left\{(X-1)^2(c_W)^{-\rho_W-1}\right\}}\right]\\ &= -\frac{\lambda_W(X-1)}{(1-\lambda_W)\rho_W}\cdot\frac{E\left[(X-1)(c_W)^{-\rho_W}\ln c_W\right]}{E\left[(X-1)^2(c_W)^{-\rho_W-1}\right]}\end{aligned} \tag{B.33}$$

Substituting equation (B.20) in (B.32):

$$\begin{aligned}\frac{\partial c_H}{\partial x_H} &= (1-\lambda_H)M_H(X-1) + \lambda_W\cdot\left[-\frac{\lambda_H M_H(X-1)}{(1-\lambda_W)}\right]\\ &= M_H(X-1)\cdot\left[(1-\lambda_H) - \frac{\lambda_W\lambda_H}{(1-\lambda_W)}\right]\\ &= \frac{M_H(1-\lambda_W-\lambda_H)(X-1)}{(1-\lambda_W)}\end{aligned} \tag{B.34}$$

Substituting equation (B.33) in (B.29):

$$\begin{aligned}
\frac{\partial G}{\partial \rho_W} &= \frac{(1-\lambda_W-\lambda_H)M_H}{1-\lambda_W} E\left[(X-1)(-\rho_H)c_H^{-\rho_H-1}\cdot\left\{-\frac{\lambda_W(X-1)}{(1-\lambda_W)\rho_W}\cdot\frac{E\left[(X-1)(c_W)^{-\rho_W}\ln c_W\right]}{E\left[(X-1)^2(c_W)^{-\rho_W-1}\right]}\right\}\right] \\
&= \frac{(1-\lambda_W-\lambda_H)\lambda_W\rho_H M_H}{(1-\lambda_W)^2\rho_W}\cdot E\left[(X-1)^2 c_H^{-\rho_H-1}\right]\cdot\frac{E\left[(X-1)(c_W)^{-\rho_W}\ln c_W\right]}{E\left[(X-1)^2(c_W)^{-\rho_W-1}\right]} \\
&= \frac{(1-\lambda_W-\lambda_H)\lambda_W\rho_H M_H}{(1-\lambda_W)^2\rho_W}\cdot\frac{E\left[(X-1)(c_W)^{-\rho_W}\ln c_W\right]\cdot E\left[(X-1)^2 c_H^{-\rho_H-1}\right]}{E\left[(X-1)^2(c_W)^{-\rho_W-1}\right]} \qquad \text{(B.35)}
\end{aligned}$$

Substituting equation (B.34) in (B.30):

$$\begin{aligned}
\frac{\partial G}{\partial x_H} &= \frac{(1-\lambda_W-\lambda_H)M_H}{1-\lambda_W}\cdot E\left[(X-1)(-\rho_H)(c_H)^{-\rho_H-1}\cdot\left\{\frac{M_H(1-\lambda_W-\lambda_H)(X-1)}{(1-\lambda_W)}\right\}\right] \\
&= -\frac{(1-\lambda_W-\lambda_H)^2 M_H^2\rho_H}{(1-\lambda_W)^2}\cdot E\left[(X-1)^2(c_H)^{-\rho_H-1}\right] \qquad \text{(B.36)}
\end{aligned}$$

Substituting equations (B.35) and (B.36) in (B.28):

$$\frac{dx_H^*}{d\rho_W} = -\frac{\frac{(1-\lambda_W-\lambda_H)\lambda_W\rho_H M_H}{(1-\lambda_W)^2\rho_W}\cdot\frac{E\left[(X-1)^2(c_W)^{-\rho_W}\ln c_W\right]\cdot E\left[(X-1)^2 c_H^{-\rho_H-1}\right]}{E\left[(X-1)^2(c_W)^{-\rho_W-1}\right]}}{-\frac{(1-\lambda_W-\lambda_H)^2 M_H^2\rho_H}{(1-\lambda_W)^2}\cdot E\left[(X-1)^2(c_H)^{-\rho_H-1}\right]}$$

Simplifying,

$$\boxed{\frac{dx_H^*}{d\rho_W} = \frac{\lambda_W}{(1-\lambda_W-\lambda_H)M_H\rho_W}\cdot\frac{E\left[(X-1)(c_W)^{-\rho_W}\ln c_W\right]}{E\left[(X-1)^2(c_W)^{-\rho_W-1}\right]}} \qquad \text{(B.37)}$$

Using the result from equation (B.23) in (B.37):

$$\frac{dx_H^*}{d\rho_W} = \frac{\lambda_W}{(1-\lambda_W-\lambda_H)M_H\rho_W}\cdot\left[-(1-\lambda_W)M_W\rho_W\right]\cdot\frac{dx_W^*}{d\rho_W} = -\frac{\lambda_W(1-\lambda_W)M_W}{(1-\lambda_W-\lambda_H)M_H}\cdot\frac{dx_W^*}{d\rho_W} \qquad \text{(B.38)}$$

Equation (B.37) implies as the degree of risk aversion of the second mover increases ($\rho_W \uparrow$) the first mover should increase their allocation into the risky asset. Further this implies that a first mover who overestimates the second mover's degree of risk aversion will overinvest in the risky asset while a first mover who underestimates the second mover's degree of risk aversion will underinvest in the risky asset.

Predictions from the model:

1. Equation (B.19) predicts that the second mover should choose safer lotteries as the first mover's choice grows progressively riskier. Thus, the *Strategy* variable for the second mover should be negative in sign.

$$\boxed{\frac{dx_W^*}{dx_H} = -\frac{\lambda_H M_H}{(1-\lambda_W)M_W}}$$

   Also as the fraction of income that the first-mover shares with the second-mover, λ_H, falls the sensitivity of the second-mover's choice to the first-mover's choice falls. The joint decision in the game artificially sets $\lambda_H = \lambda_W = 0.5$. However, if player's lived experience in the household influences their choices in the game the actual value of λ_H may be different. In our context, where men are in control of most household financial resources while most women do not possess independent income sources it is possible that the actual value of λ_H for men is closer to zero than for women.

2. Equation (B.37) implies that first movers who overestimate the second mover's degree of risk aversion overinvest in the risky asset whereas first mover who underestimate the second mover's degree of risk aversion underinvest in the risky asset.

B.6 Addendum: Aligning with First Mover

While the first mover does not know what their spouse as second mover chooses, the second mover does know what their spouse's choice is. This knowledge can affect how second movers choose especially women. There is evidence that in patriarchal societies, women are more likely than men to defer to their spouses in decisions (**?**). To reflect this behavior, I modify the second mover's utility function as follows:

$$U_W(c_W, x_H, x_W) = \frac{c_W^{1-\rho_W}}{1-\rho_W} - \delta(x_H - x_W)^2 \tag{B.39}$$

where δ is a non-negative constant reflecting to what degree the second mover "dislikes" diverging from the first mover's choices.

B.6.1 Second Mover's New Optimization Problem

The second mover maximizes:

$$\max_{x_W \in [0,1]} \quad E\left[U_W(y_W, y_H)\right] = E\left[\frac{((1-\lambda_W)y_W + \lambda_H y_H)^{1-\rho_W}}{1-\rho_W} - \delta(x_H - x_W)^2\right]. \tag{B.40}$$

The first-order condition is:

$$\begin{aligned}
&\frac{\partial}{\partial x_W}\left[E\left\{\frac{c_W^{1-\rho_W}}{1-\rho_W}\right\}-\delta(x_H-x_W)^2\right]=0\\
\implies\ & E\left[(c_W)^{-\rho_W}(1-\lambda_W)M_W(X-1)\right]+2\delta(x_H-x_W)=0\\
\implies\ & (1-\lambda_W)M_W E\left[(c_W)^{-\rho_W}(X-1)\right]+2\delta(x_H-x_W)=0
\end{aligned}\tag{B.41}$$

This defines $x_W^*(x_H)$, the wife's best response and so also $y_W^*(x_W^*(x_H))$.

$$y_W^*(x_H)=x_W^*(x_H)M_WX+(1-x_W^*(x_H))M_W \tag{B.42}$$

$$\implies \frac{dy_W^*(x_H)}{dx_H}=M_W(X-1)\frac{dx_W^*(x_H)}{dx_H} \tag{B.43}$$

B.6.2 IFT to find $\frac{dx_W^*}{dx_H}$

Rewriting F.O.C. in equation (B.41) as:

$$F(x_W,x_H)=(1-\lambda_W)M_WE\left[(c_W)^{-\rho_W}(X-1)\right]+2\delta(x_H-x_W)=0 \tag{B.44}$$

Using IFT:

$$\frac{dx_W^*}{dx_H}=-\frac{\frac{\partial F}{\partial x_H}}{\frac{\partial F}{\partial x_W}} \tag{B.45}$$

$$\frac{\partial F}{\partial x_H}=-(1-\lambda_W)M_W\cdot E[\rho_W(c_W)^{-\rho_W-1}(X-1)\frac{\partial c_W}{\partial x_H}]+2\delta \tag{B.46}$$

$$\frac{\partial F}{\partial x_W}=-(1-\lambda_W)M_WE[\rho_W(c_W)^{-\rho_W-1}(X-1)\frac{\partial c_W}{\partial x_W}]-2\delta \tag{B.47}$$

Substituting equation (B.15) in (B.46) and equation (B.16) in (B.47),

$$\frac{\partial F}{\partial x_H}=-(1-\lambda_W)\lambda_HM_WM_H\rho_WE[(c_W)^{-\rho_W-1}(X-1)^2]+2\delta \tag{B.48}$$

$$\frac{\partial F}{\partial x_W}=-(1-\lambda_W)^2M_W^2\rho_WE[(c_W)^{-\rho_W-1}(X-1)^2]-2\delta \tag{B.49}$$

From equations (B.11), (B.48), and (B.49),

$$
\begin{aligned}
\frac{dx_W^*}{dx_H} &= -\frac{-(1-\lambda_W)\lambda_H M_W M_H \rho_W \cdot E[(c_W)^{-\rho_W - 1}(X-1)^2] + 2\delta}{-(1-\lambda_W)^2 M_W^2 \rho_W \cdot E[(c_W)^{-\rho_W-1}(X-1)^2] - 2\delta} \\
&= -\frac{(1-\lambda_W)\lambda_H M_W M_H \rho_W \cdot E[(c_W)^{-\rho_W - 1}(X-1)^2] - 2\delta}{(1-\lambda_W)^2 M_W^2 \rho_W \cdot E[(c_W)^{-\rho_W-1}(X-1)^2] + 2\delta}
\end{aligned} \tag{B.50}
$$

Rearranging terms in equation (B.50) produces:

$$
\boxed{\left.\frac{dx_W^*}{dx_H}\right|_{\delta>0} = -\frac{\lambda_H M_H - \frac{2\delta}{(1-\lambda_W)M_W\rho_W \cdot E[(c_W)^{-\rho_W-1}(X-1)^2]}}{(1-\lambda_W)M_W + \frac{2\delta}{(1-\lambda_W)M_W\rho_W \cdot E[(c_W)^{-\rho_W-1}(X-1)^2]}}} \tag{B.51}
$$

Now under the simple case where second movers do not care whether their choices diverge from their spouse's,

$$
\boxed{\left.\frac{dx_W^*}{dx_H}\right|_{\delta=0} = -\frac{\lambda_H M_H}{(1-\lambda_W)M_W}} \tag{B.52}
$$

$\left.\frac{dx_W^*}{dx_H}\right|_{\delta>0}$ adds a positive amount to the numerator and subtracts a positive amount from the denominator of $\left.\frac{dx_W^*}{dx_H}\right|_{\delta=0}$. Thus,

$$
\left.\frac{dx_W^*}{dx_H}\right|_{\delta>0} \geq \left.\frac{dx_W^*}{dx_H}\right|_{\delta=0}
$$

for $\delta \geq 0$.

Predictions from the modified model for second-mover decision-making:

1. Equation B.51 suggests that if second movers place some positive value to aligning with their spouse's choices then $\left.\frac{dx_W^*}{dx_H}\right|_{\delta>0}$ will be less negative than otherwise. If δ is large enough $\left.\frac{dx_W^*}{dx_H}\right|_{\delta>0}$ could be positive. If women care more about align with their spouse than men then then the *Strategy* variable for women should be more positive than for men second movers.

Appendix C: Tables & Figures

Table B.1: Difference in characteristics of men participants and non-participants

	Participants		Non-participants		(1) - (2)	
	Mean	Std. Dev.	Mean	Std. Dev.	Diff.	Std. Err.
	(1)		(2)		(3)	
Age (years)	42.69	(10.93)	42.93	(14.40)	-0.24	(0.96)
Education:						
Less than primary (%)	15.03	(35.76)	21.84	(41.44)	-6.81**	(3.05)
Primary (%)	32.54	(46.88)	27.59	(44.82)	4.96	(3.86)
Secondary (%)	34.12	(47.44)	36.21	(48.20)	-2.08	(3.94)
Higher-secondary or more (%)	18.30	(38.69)	14.37	(35.18)	3.94	(3.16)
Number of assets:						
Mobile phone	2.42	(1.10)	2.33	(1.12)	0.09	(0.09)
Television	0.78	(0.55)	0.70	(0.55)	0.08*	(0.05)
Cow/Buffalo	1.22	(1.53)	0.92	(1.49)	0.30**	(0.13)
Poultry	15.63	(14.50)	15.00	(15.31)	0.63	(1.21)
Observations	885		174			

Notes: Standard deviation in parentheses for mean outcomes. Differences and associated significance levels were derived from two sample t-tests. Standard errors for the t-tests are presented in parentheses. * $p<0.1$, ** $p<0.05$, and *** $p<0.01$.

Table B.2: Results from regression of the difference in number of livestock assets reported by women from their husband's report on the errors made by men and women in *Joint* game.

	(1)	(2)
Wife: *Livestock* - Husband: *Livestock*		
Wife *Error*	0.466**	0.439**
	(0.185)	(0.183)
Husband *Error*	0.077	0.107
	(0.200)	(0.198)
Constant	-2.331***	-2.565
	(0.768)	(3.926)
Controls	No	Yes
R-squared	0.008	0.082
Observations	849	849

Notes: Husbands and wives were each asked to report the number of cow/buffalo, goat/sheep, and poultry owned by the household. The dependent variable is created by differencing the number reported by the husband from the number reported by his wife. Controls include risk preferences, age, education, and occupation of player and spouse, family composition, women's decision-making powers and wealth index. Sample excludes households where both partners reported not having any livestock. Standard errors reported in parentheses. * p<0.1, ** p<0.05, and *** p<0.01.

Table B.3: Average Big Five Personality Traits by Gender

	Men (1)	Women (2)	Difference (2) - (1)
Conscientiousness	5.769	5.763	-0.006
			(0.053)
Extraversion	4.621	4.475	-0.147***
			(0.042)
Agreeableness	5.597	5.385	-0.212***
			(0.048)
Openness	4.792	4.699	-0.093
			(0.065)
Neuroticism	3.062	3.492	0.429***
			(0.059)
Observations	885	885	1770

Notes: Difference column reports the coefficient on the Female indicator from regressions of each trait on a female dummy. Standard errors in parentheses. Personality traits are measured on a scale ranging from 1 to 7. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure B.1: Visuals presented to players to describe the lottery options.

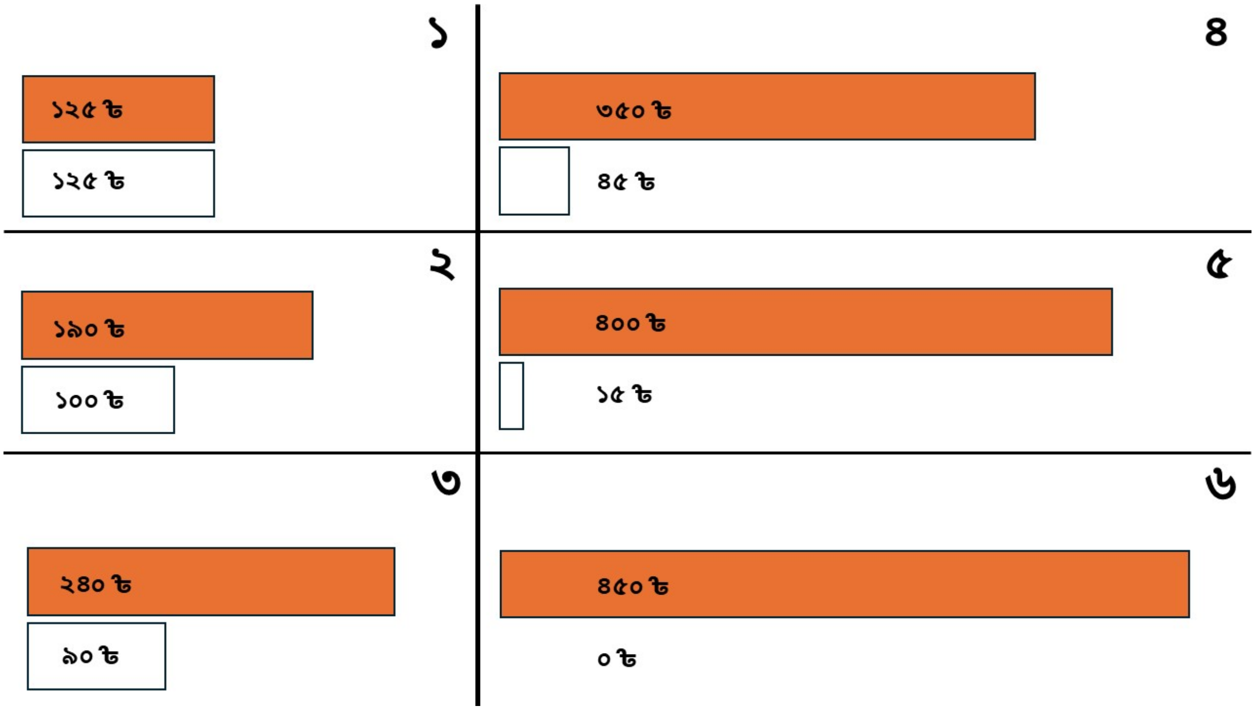
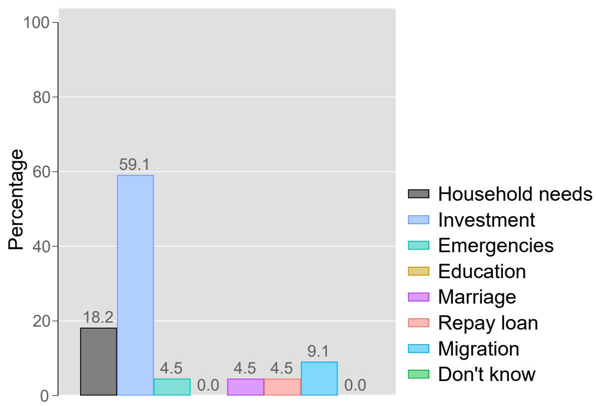
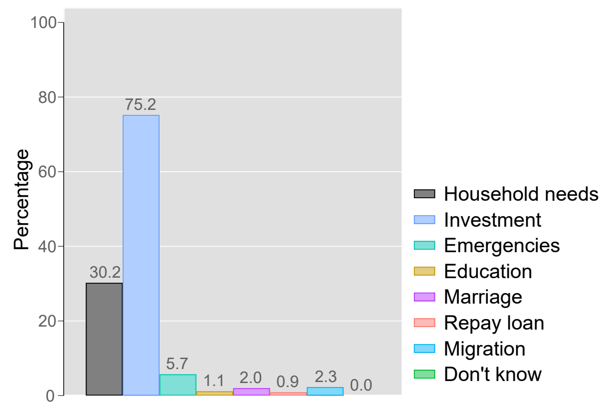


Figure B.2: Reported purpose for obtaining loan of men and women.

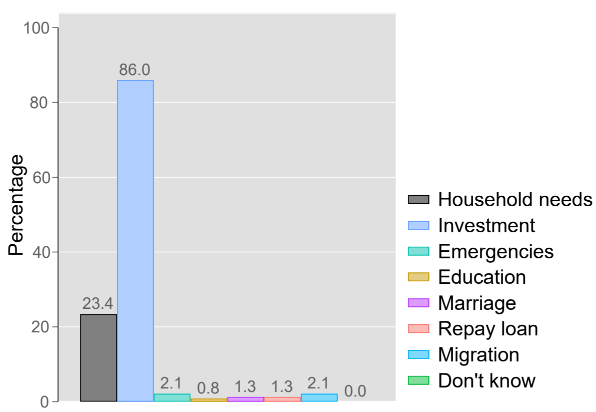
(a) Women's loans: Self reported



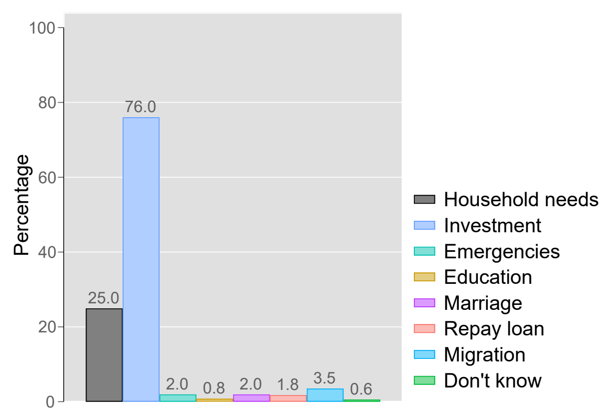
(b) Women's loans: Spouse reported



(c) Men's loans: Self reported



(d) Men's loans: Spouse reported



Notes: Respondents were asked to report how many active loans the household had in the past 12 months, the primary decision-maker for each loan, the purpose for each loan, and the source from which they were obtained. If a person reported the household having any, they were asked the top two reasons for taking the loan.